

Chapter 1

Oleo-hydraulic and pneumatic technologies



Universidad de Oviedo
Área de Mecánica de Fluidos

Escuela Politécnica de Ingeniería (EPI) de Gijón
4º curso – Grado en Ingeniería Mecánica

**FLUID POWER: HYDRAULIC AND PNEUMATIC
TECHNOLOGIES**

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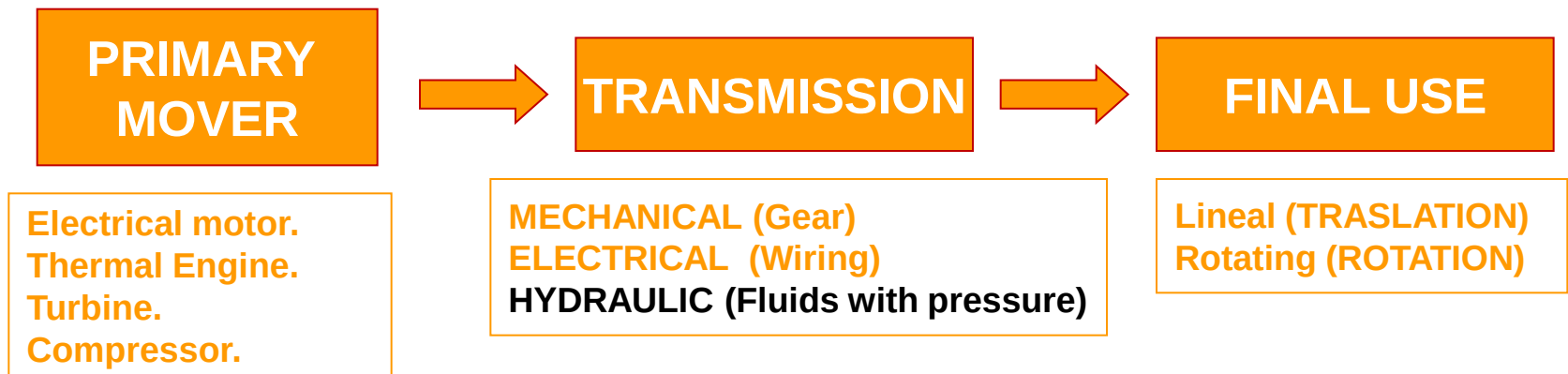
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1.1. Power transmission systems.

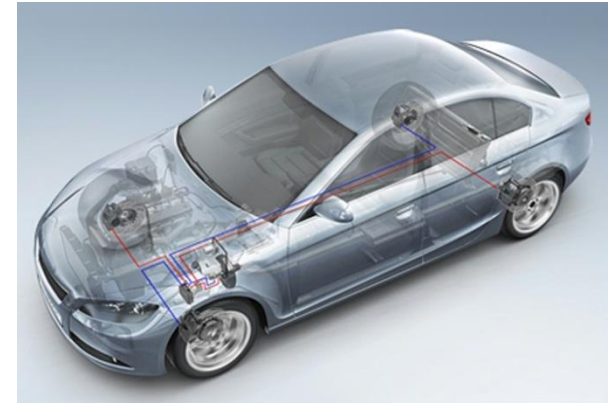
Technological scope:

- ❑ It establishes a natural link between a **primary mover** and final mechanism where a **useful mechanical work** (force or torque) is obtained. The whole procedure is often named as fluid power transmission or, simply, **fluid power**.
- ❑ **Oleo-hydraulic** promotes the power transmission using an incompressible fluid (often **oil**).
- ❑ The **Pneumatic** studies the power transmission (o energy transmission) using a gas (**air**, almost always) searches the same goal.



1.1. Power transmission systems.

Application ranges:



❑ **Mechanical:** the elder systems, practical and economical in some applications, but heavy, too complex and expensive in other. Small lengths **(around one meter)**.

❑ **Electrical:** more recent, simpler and more flexible than the mechanical ones. Long lengths **(around one kilometer)**.

❑ **Fluid power:** intermediate between both previous ones, with a slower development, their interest grew due to the aeronautical applications basically due to their good relationship power/weight. Intermediate distances **(around ten meters)**.

1.1. Power transmission systems.

Power transmission characteristics:

	ELECTRIC TRANSMISSION	HYDRAULIC TRANSMISSION	PNEUMATIC TRANSMISSION
LEAKS		Contamination (oils)	Without risks (energy losses)
AMBIENTAL CONSIDERATIONS	Explosion risks, independent of temperature variations	Sensible to temperature changes. Danger of contamination and explosion	Not affected by temperature changes, almost no risk of explosion
ENERGY STORAGE	Difficult, only for small batteries	Limited, with the help of gases	Easy
ENERGY TRANSMISSION	Without limits, with some energy losses	Up to 100 m. Velocities between 2 and 6 m/s	Up to 1000 m. Velocities between 20 and 40 m/s
OPERATION VELOCITIES		0.5 m/s	1.5 m/s
COSTS	Low	High	Very high
LINEAR MOVEMENT	Difficult and expensive. Only for small forces and very costly regulation	Very simple with linear actuators. Optimum control speeds and high forces	Very simple with linear actuators. Speed dependent to adjust velocities
ROTATING MOVEMENT	Easy and with high power	Easy for high torques and low rotational speeds	Easy, not so high efficiency for high speeds
POSITIONING	High precision	Precision might be reached	Average precision (very load dependent)
STABILITY	High with mechanical unions	Very high, due to oil incompressibility	Low, due to air compressibility
FORCES	High loads, but difficult to handled over-loads	Protected towards over- loads due to high forces	Protected against overloads, but for small forces



1.2. Fluid power systems.

Advantages of systems with pressurized fluid:

- ❑ **Easy control and positioning with precision.** (Tolerances till 1/10000 in)
- ❑ **Multiplication of forces.** (From a few grams to hundred of tons)
- ❑ **Constant force/torque supply.** (Unique system able to do it)
- ❑ **Simple, safe and economical systems.** (Less moving elements)
- ❑ **Allow linear/rotating movements**
- ❑ **Additionally** (auto-lubrication, overload protection, speed regulation, energy storage...)

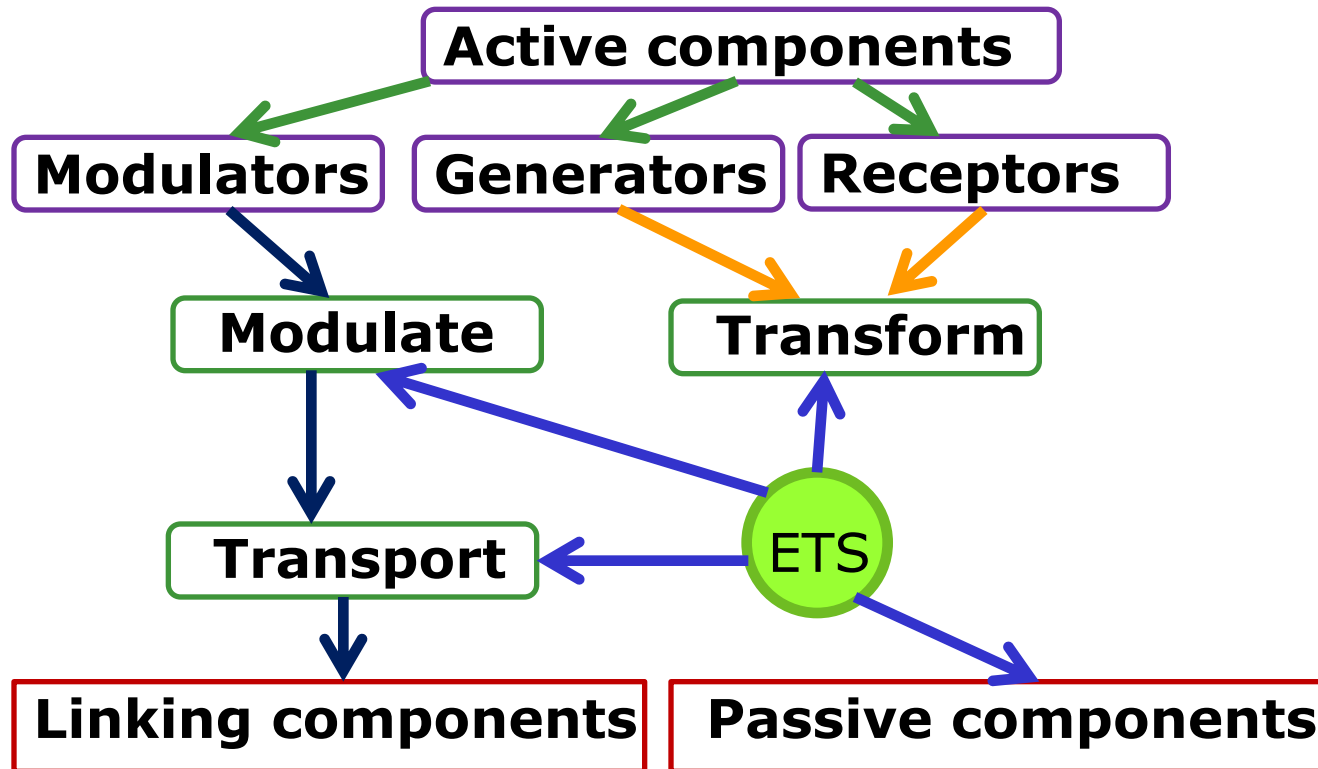
Disadvantages:

- ❑ Contamination issues because of oil leaks
- ❑ Temperature limitations, minimum and maximum (oil viscosity)
- ❑ Filtering necessity
- ❑ Explosion in case of improper designed systems

The systems with pressurized fluid have the highest Power/Weight ratio of all power transmission systems known

1.2. Fluid power systems.

Components in the power/energy transmission.



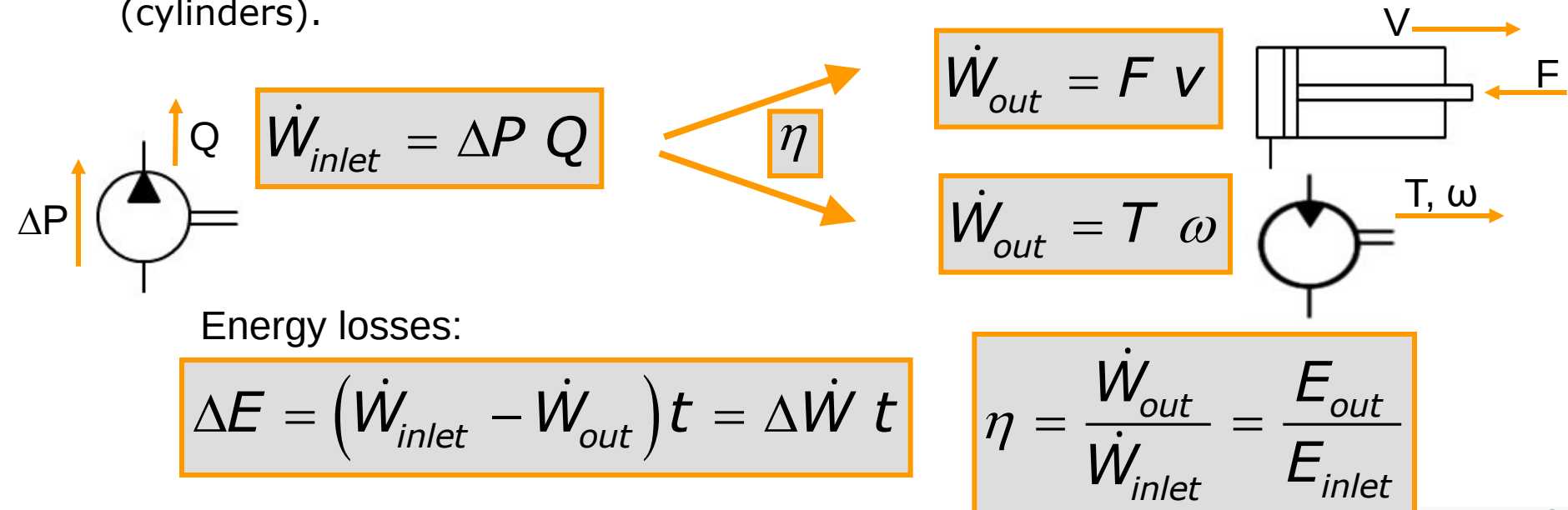
□ **Fundamental equation:** **Power transmission (or energy):**

$$\dot{W} = \Delta P Q$$

1.2. Fluid power systems.

Active components:

- ❑ **Generators:** produce an energy increase in the fluid (pressure). Basically, pumps and compressors. Their main working parameter is the cubic capacity.
- ❑ **Modulators:** they promote the energy adaptation to the useful effect intended in the transmission. They produce an inherent energy-head loss.
- ❑ **Receptors:** they collect the energy at its final part of the transmission and promote the useful effect searched. Hydraulic motors and actuators (cylinders).





1.2. Fluid power systems.

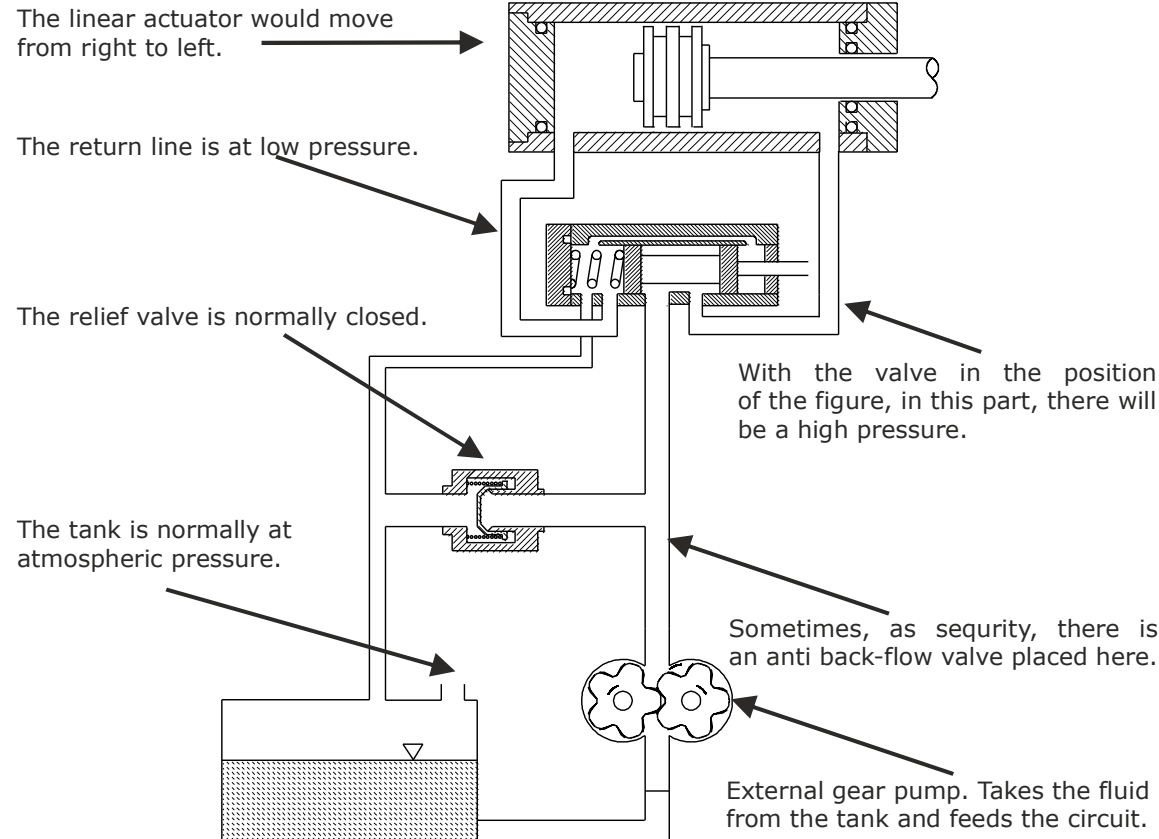
Linking components:

- ❑ **Fluid:** it transports the energy (power), serving as lubrication for the mobile parts, heat exchanger with the different parts of the circuit, it must be filtered and sometimes it has some additives.
- ❑ **Pipes or conducts:** they transport the fluid, their choosing, design and installation have an effect on the efficiency and reliability of a given fluid pressure circuit.

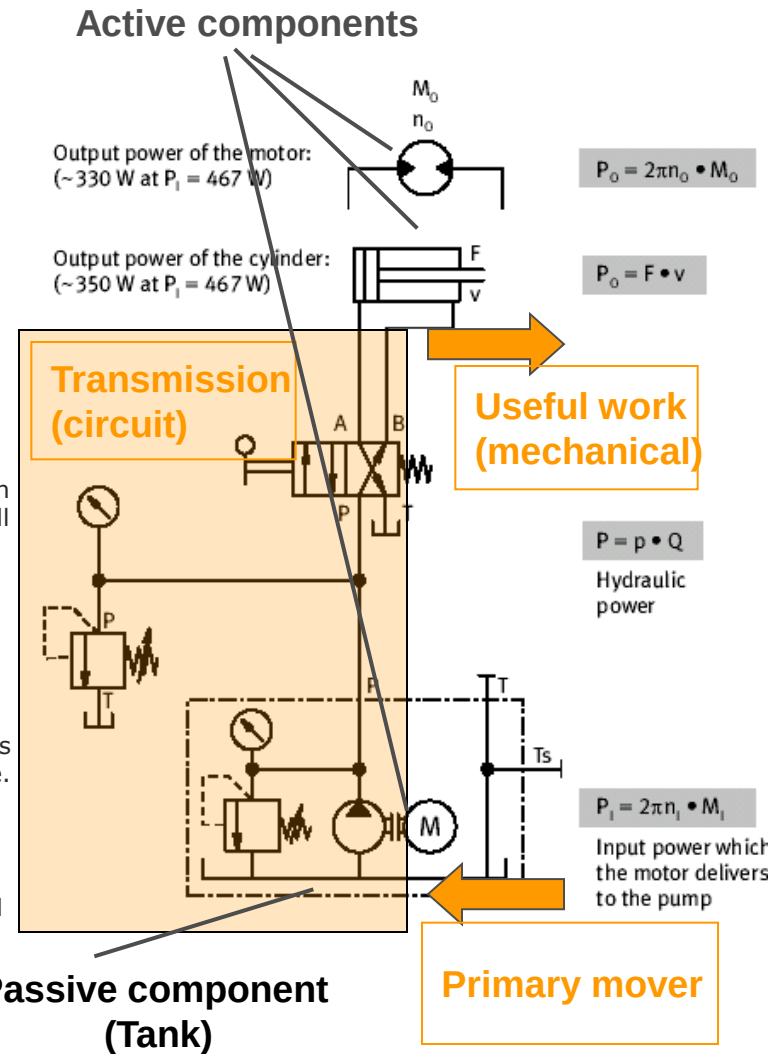
Passive components:

- ❑ **Tank:** it keeps the fluid while doing some other functions: basic filtering, heat exchanger, promote the fluid to be without any movement...
- ❑ **Heat exchangers:** in most of the applications, they cool down the fluid, but they also might be used as heaters (for outdoor installations, for instance).
- ❑ **Filters and lubrication.**
- ❑ **Silencers:** In pneumatic installations, the atmosphere outlets often include them.

1.2. Fluid power systems.



Sketch representation



Symbol representation

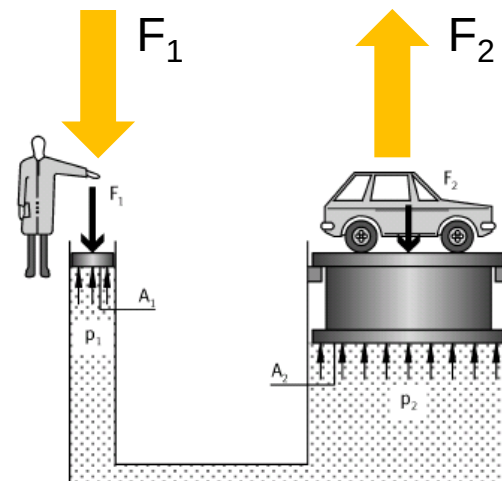
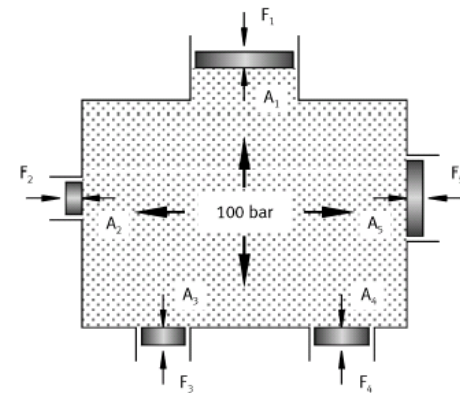
1.2. Fluid power systems.

Pressure concept. ($p=F/S$)

- ❑ **Power transmission:** When the force **F** acts on a given surface **A** in a closed system (**hydrostatics**), a pressure **P** is generated and affects the whole system (**Pascal's principle**). With a high pressure (typical in **oleo-hydraulics**), the effect of the liquid column is negligible.
- ❑ **Power transmission (ideal):** As the pressure is transmitted to all the fluid points, it is possible to transmit forces, generating big forces (very slow) using a small force.

$$p_1 = p_2 \Rightarrow \frac{F_1}{A_1} = \frac{F_2}{A_2} \Rightarrow F_1 = \frac{A_1}{A_2} F_2$$

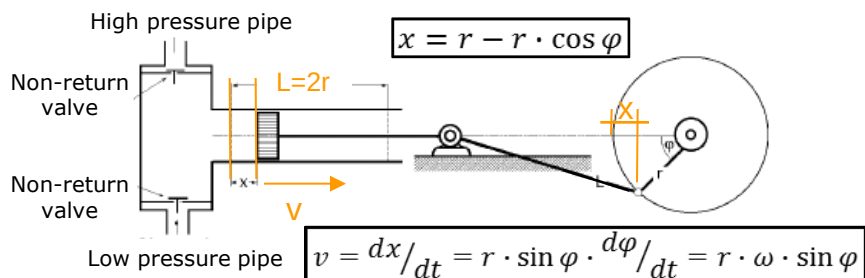
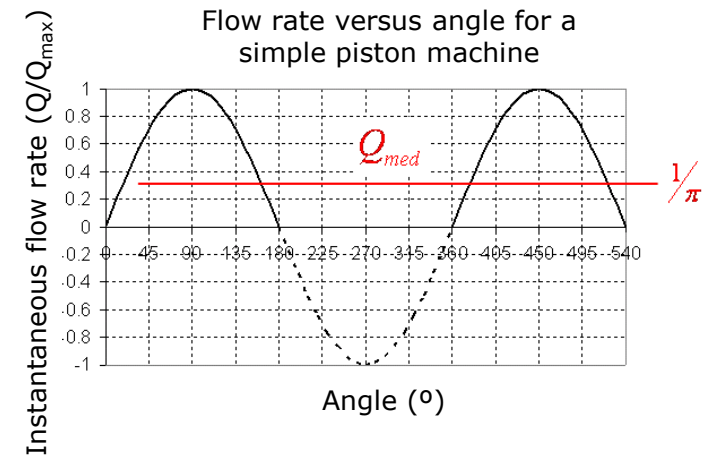
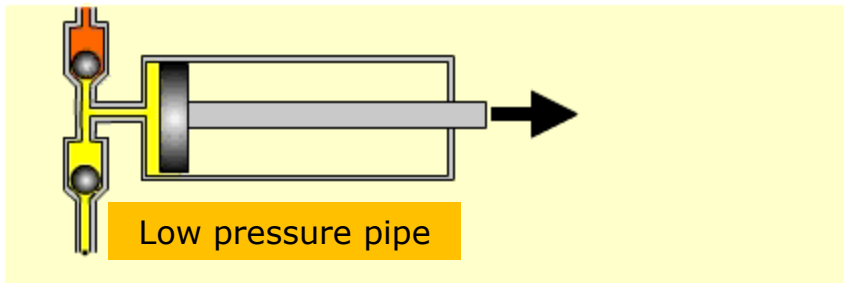
$$\dot{W}_1 = \dot{W}_2 \Rightarrow F_1 v_1 = F_2 v_2 \Rightarrow v_2 = \frac{A_1}{A_2} v_1$$



1.2. Fluid power systems.

Flow rate concept.

- ❑ The positive displacement machines generate a flow using the effect of creating and destroying **confined volumes**.
- ❑ Therefore, the flow rate is defined as the generated volume by time unit. The generated volume in a full revolution of the shaft is widely known as **cubic capacity or volumetric displacement (cm^3/rev)**.
- ❑ **Simple effect piston.**



$$d\vartheta = S_B \cdot dr = S_B \cdot r \cdot \sin \varphi d\varphi \quad [\text{cm}^3]$$

$$\vartheta = \int_0^\pi S_B \cdot r \cdot \sin \varphi d\varphi = 2 \cdot S_B \cdot r = S_B \cdot L \quad [\text{cm}^3/\text{rev}]$$

$$Q_{th} = \vartheta \cdot n = S_B \cdot L \cdot n \quad [\text{cm}^3/\text{min}]$$



1.2. Fluid power systems.

Hydraulic power:

$$\dot{W}_{fluid} = \Delta P Q$$

- ❑ It is obtained due to a **high pressure**, as the flow rate value is quite low: **compact elements and small diameters in the pipes**.
 - ❑ In **oleo-hydraulics**, it becomes of the order of 100 **bar**: 150 bar = 15 MPa = $150 \cdot 10^5$ Pa
 - ❑ **1 bar = 0.1 MPa $\approx$ 1 kg/cm²**
 - ❑ For **Pneumatics**, the typical value is 7 bar.
- ❑ The flow rate is expressed in **cm³/s** or in **l/min**
 - ❑ For pumps and motors: $Q = \vartheta n$ (remark n expressed in rev/min and ϑ is the cubic capacity in cm³/rev).
 - ❑ For linear actuators: $Q = v A$
- ❑ Sometimes, the IS of units is not the best choice for flow rates and pressures.
- ❑ But certainly for power units, but remark: **1 bar x 1 l/min = 1.67 W**



1.3. Comparison between Oleo-hydraulic and Pneumatic systems.

Dynamic factors (forces).

Force, Torque ↑

	Suitable for	Less suitable for
Oleo-hydraulics	- High torques and low speeds. - High forces and torques in the starting up protocols. - Frequent changes in movement directions. - Low inertia.	- High speeds.
Pneumatics	- High speeds during long times.	- Precise movements. - High torques at low speed.

Speed ↓





1.3. Comparison between Oleo-hydraulic and Pneumatic systems.

Operative factors (speed) and security.

	Suitable for	Less suitable for
Oleo-hydraulics	- Lineal alternative movements and big loads. - Variable speeds. - Be able to stop speeding ups. - The control might be of any possible kind. - If the load is variable, it should keep a high efficiency.	
Pneumatics	- Variable speed. - Overloads, vibrations, shocks. - The control might be of any possible kind.	- Lineal alternative movements with big loads.

+ Simplicity
↓

	Suitable due to	Less suitable due to
Oleo-hydraulics	- Less dangerous for the operators.	- Formation needs, problem solution.
Pneumatics	- Formation needs.	- More dangerous for the operator.

+ Dangerous
↓



1.3. Comparison between Oleo-hydraulic and Pneumatic systems.

Economical and environmental factors.

+ Simplicity ↓

	Suitable due to	Less suitable due to
Oleo-hydraulics	- Low installation costs if the power is high and the load and speed could change. - Robustness.	- Maintenance.
Pneumatics	- Maintenance.	- High installation costs for high power.

Cost + ↓

+ Waste, Particles ↑

	Suitable due to or for	Less suitable due to or for
Oleo-hydraulics	- Low quotient weight/power. - Rarefied atmosphere. - Liquid immersion.	- It leaves particles. - Explosion risks. - Relationship between components initially independent. - Extreme temperatures.
Pneumatics	- High temperatures. - Rarefied atmosphere.	- Low quotient weight/power. - Low temperatures. - Noisy.

Noise + ↓



1.3. Comparison between Oleo-hydraulic and Pneumatic systems.

COMPARATIVE

	Oleo-hydraulics	Pneumatics
	Constant flow rate	Constant pressure
<i>Forces</i>	Overload resistance. Very high loads. More pressure up to 700 bar.	Overload resistance. Not too high loads. Less pressure, up to 17 bar.
<i>Lineal movement</i>	Easy. Feasible control of both position and speed.	Easy. Speeds depend upon the load.
<i>Angular movement</i>	Easy. High torque. Low speed.	Easy. Low efficiency. High speeds.
<i>Energy accumulation</i>	Among certain limits.	Easy.
<i>Environment</i>	Sensitive to temperature and dirt variations. Dangerous in case of fire, due to the leaks. Oil contamination. Closed circuit.	It does not produce explosions. Not sensitive to temperature variations. Open circuit: noisy.



1.3. Comparison between Oleo-hydraulic and Pneumatic systems.

COMPARATIVE

<i>Relative cost of the transmitted energy</i>	1	2.5
<i>Stability</i>	Good, because oil is an incompressible fluid.	Low, because air is a compressible fluid.
<i>Precise positioning</i>	Up to 1 μm .	Up to 0.1 mm.
<i>Fluid</i>	It uses a liquid with higher inertia, higher losses and more expensive. Conditioning: cooling, heating, filtering, additives.	It uses air: cheap and safe. Conditioning: humidity elimination, dust, oil adding.

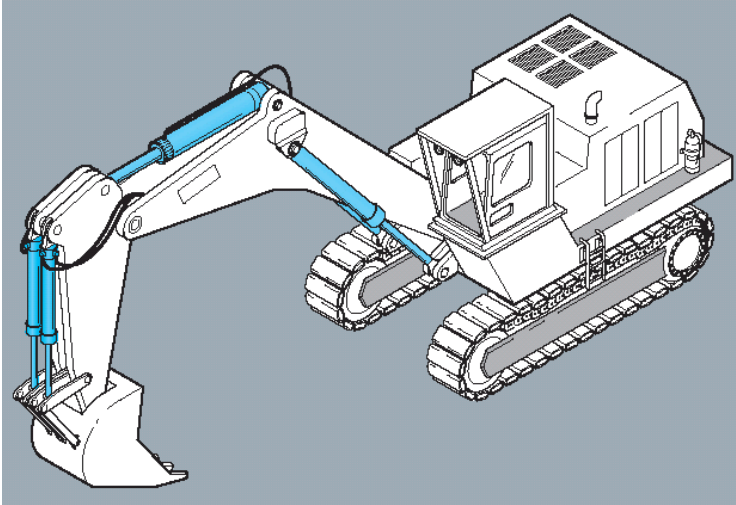
1.4. Applications.

Typical Oleo-hydraulic applications. Civil works – cranes.



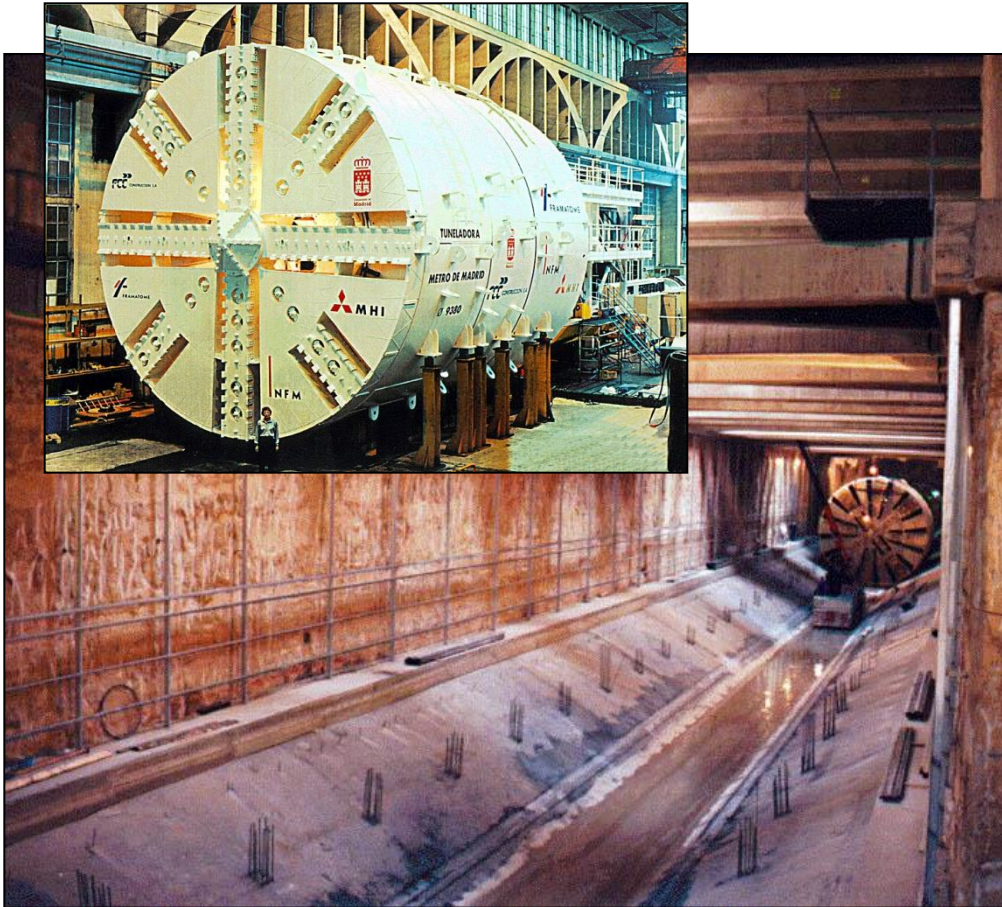
1.4. Applications.

Typical Oleo-hydraulic applications. Civil works – excavators.



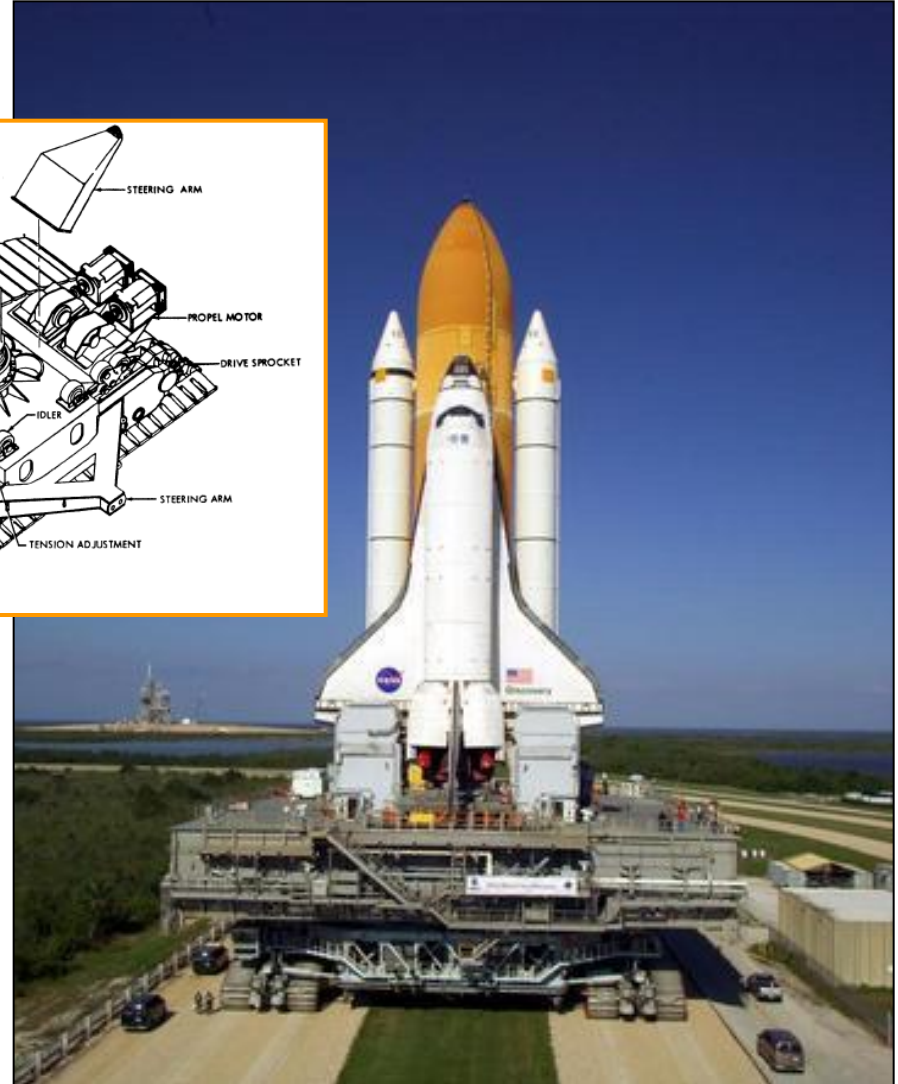
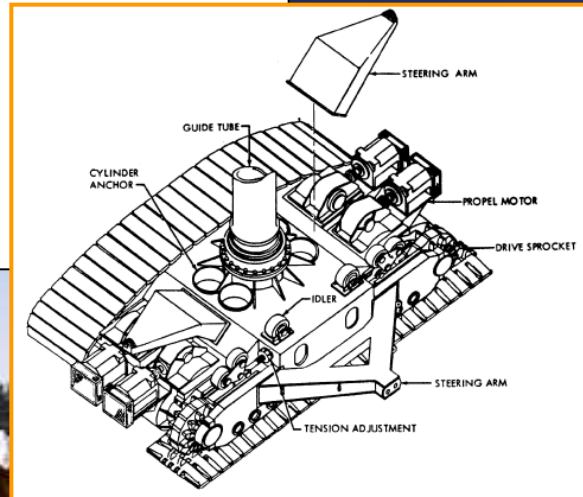
1.4. Applications.

Typical Oleo-hydraulic applications. Civil works – tunnelling.



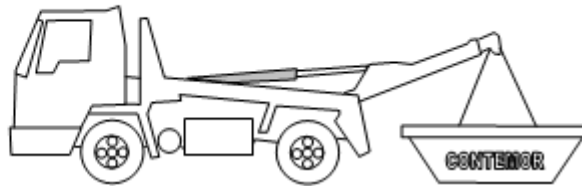
1.4. Applications.

Typical Oleo-hydraulic applications. Transport – caterpillars.



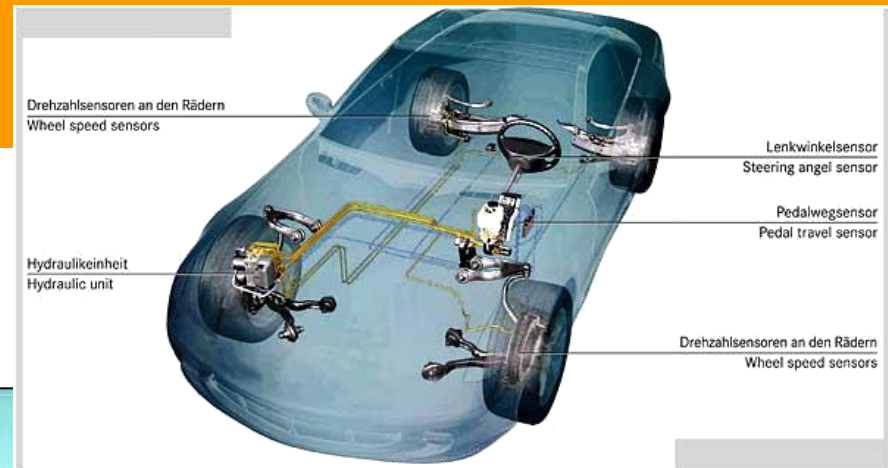
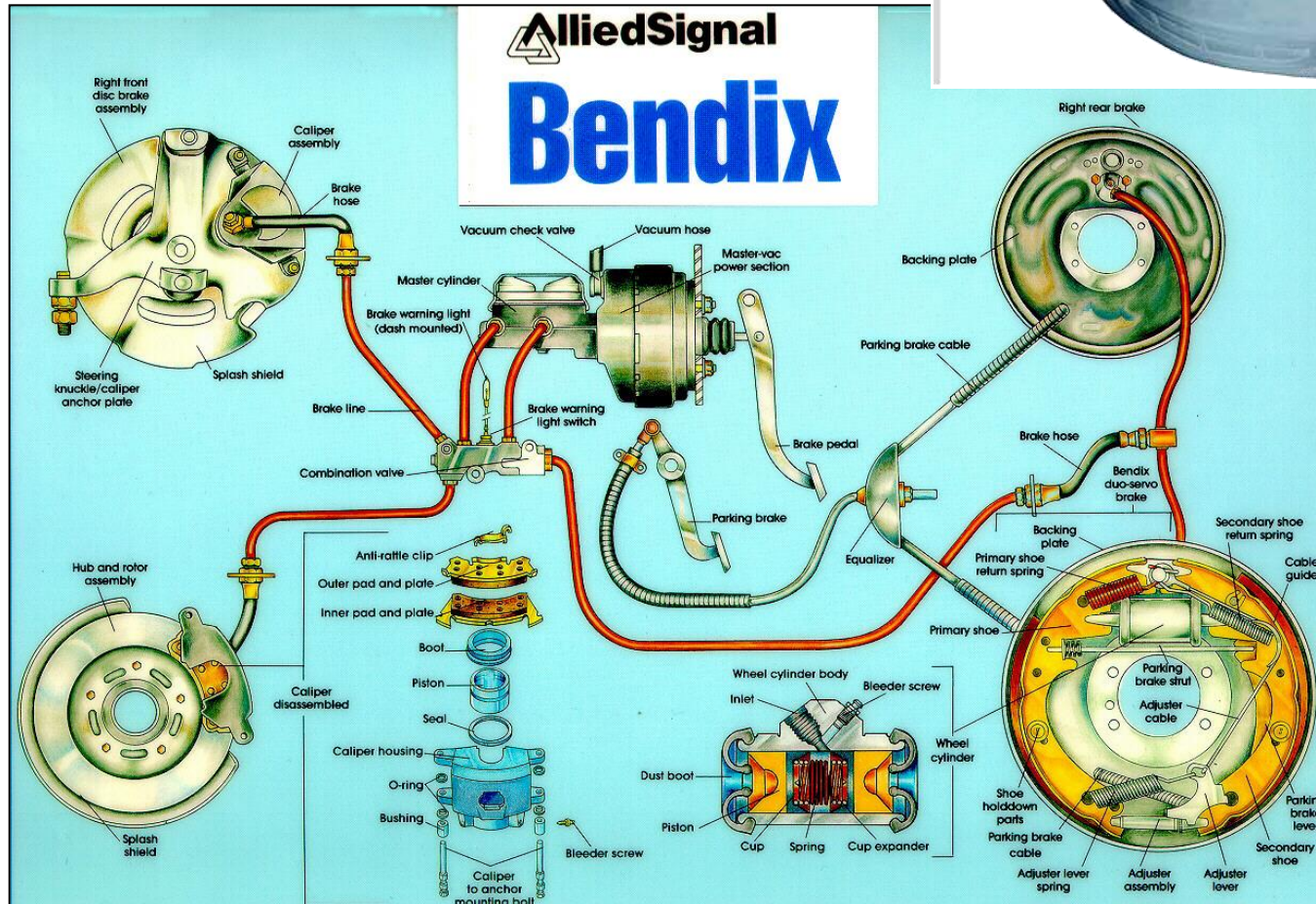
1.4. Applications.

Oleo-hydraulic applications: transport – dump trucks.



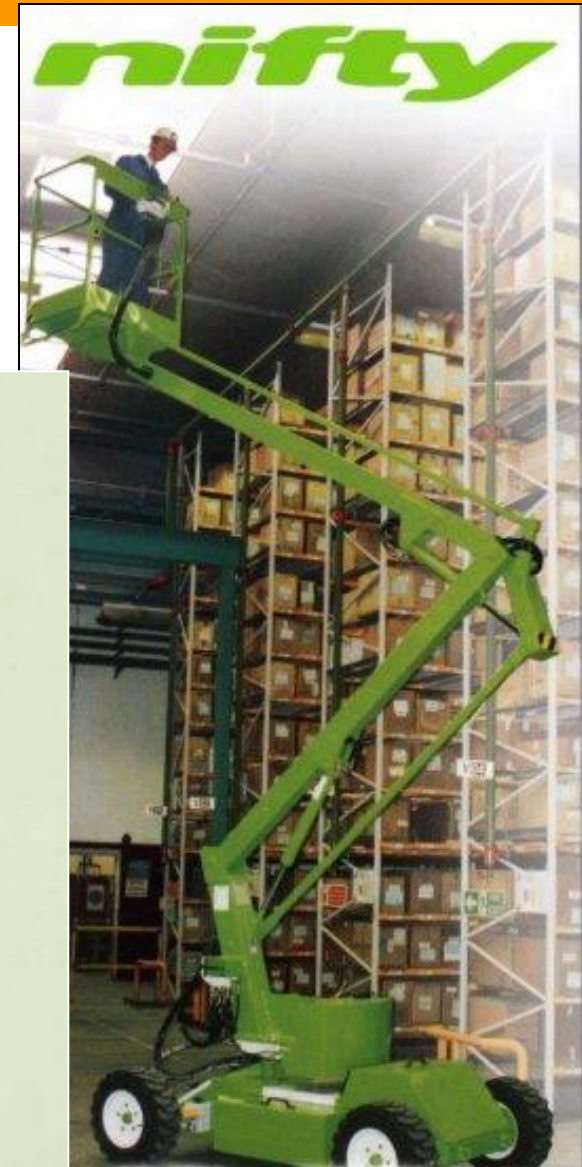
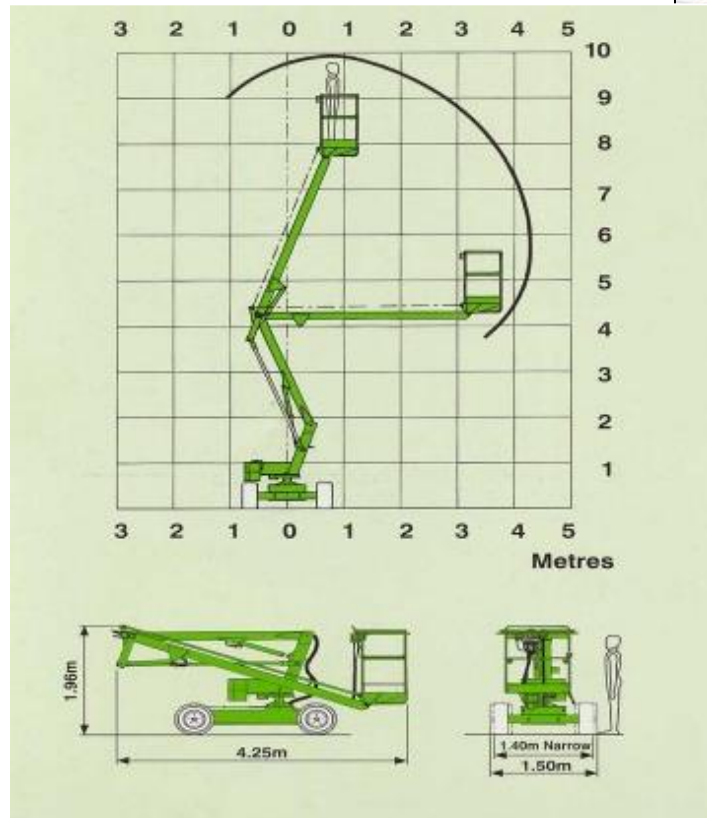
1.4. Applications.

Oleo-hydraulic applications: transport – brakes.



1.4. Applications.

Oleo-hydraulic applications: transport – platforms.



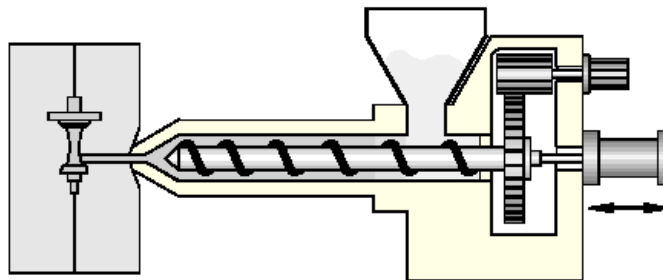
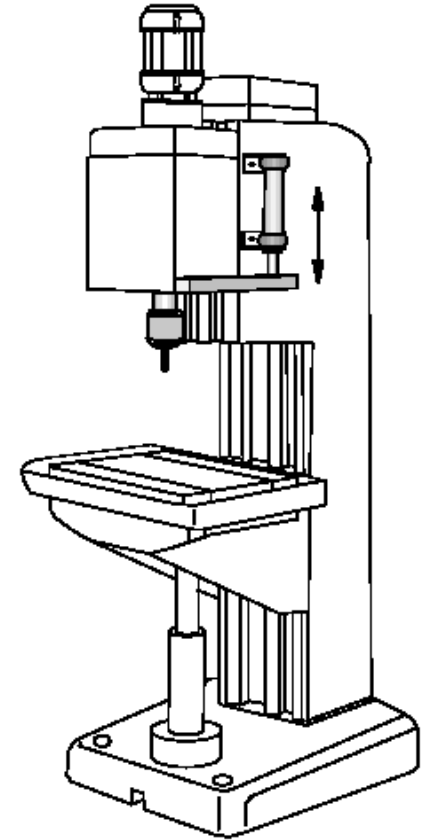
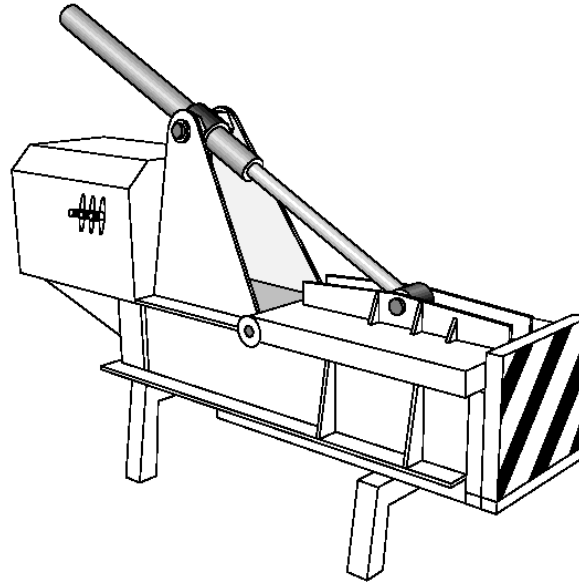
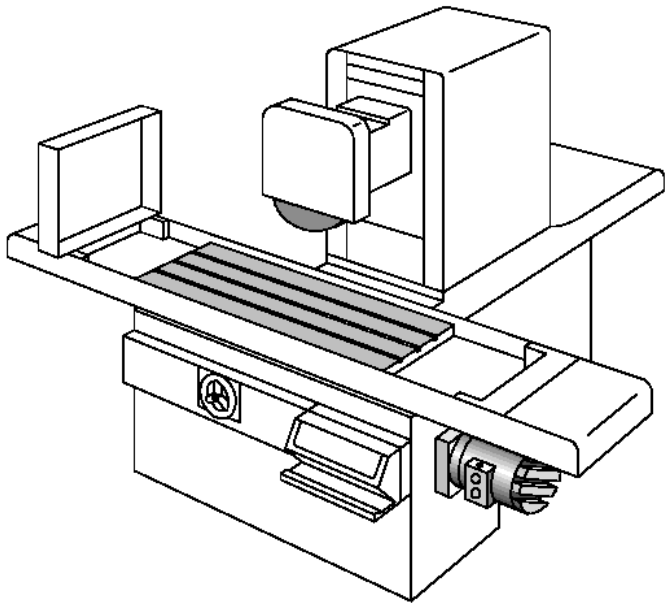
1.4. Applications.

Oleo-hydraulic applications: Civil works – bridge building.



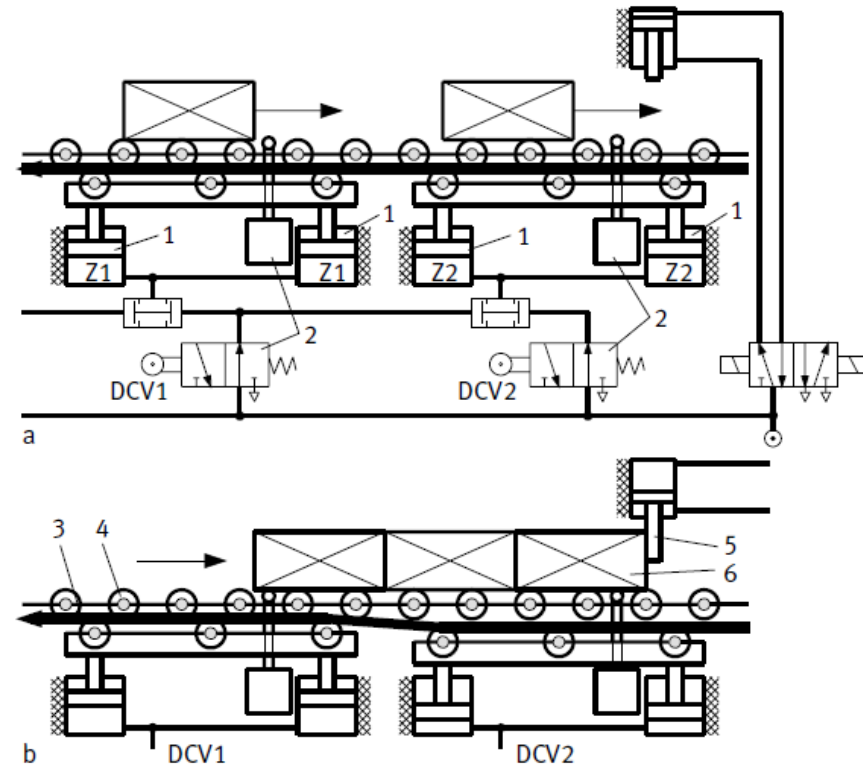
1.4. Applications.

Oleo-hydraulic applications: machinery – mechanical tools.



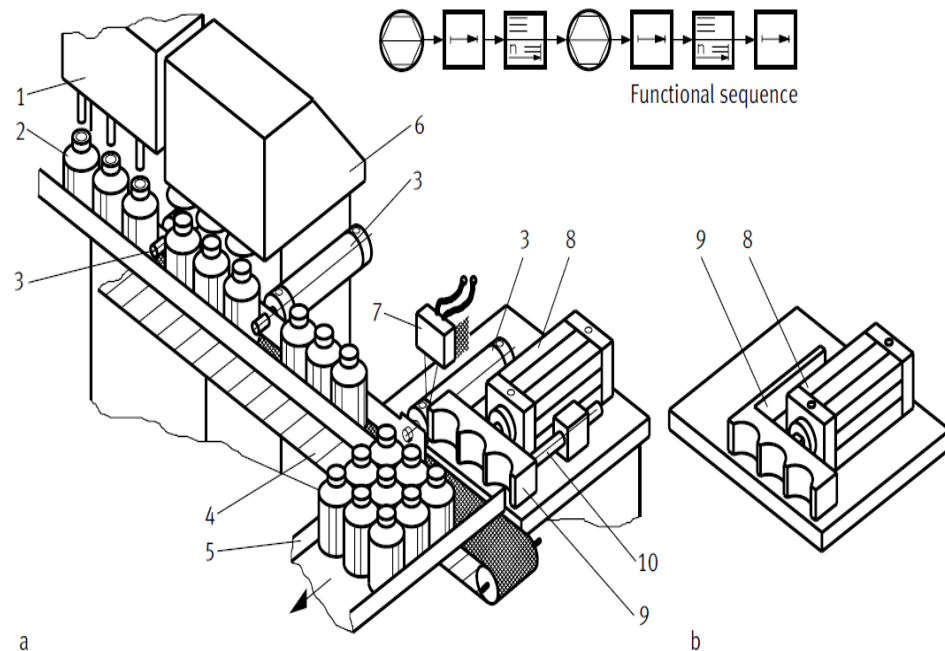
1.4. Applications.

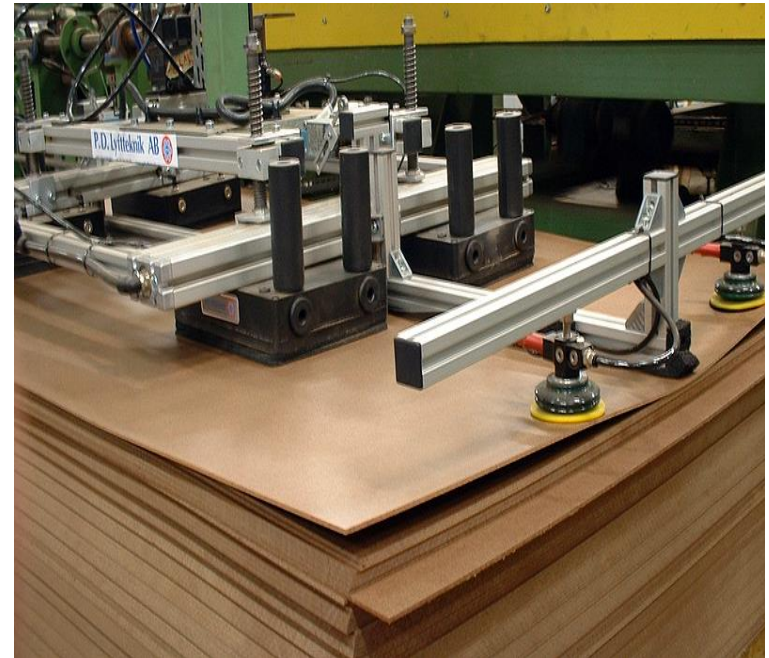
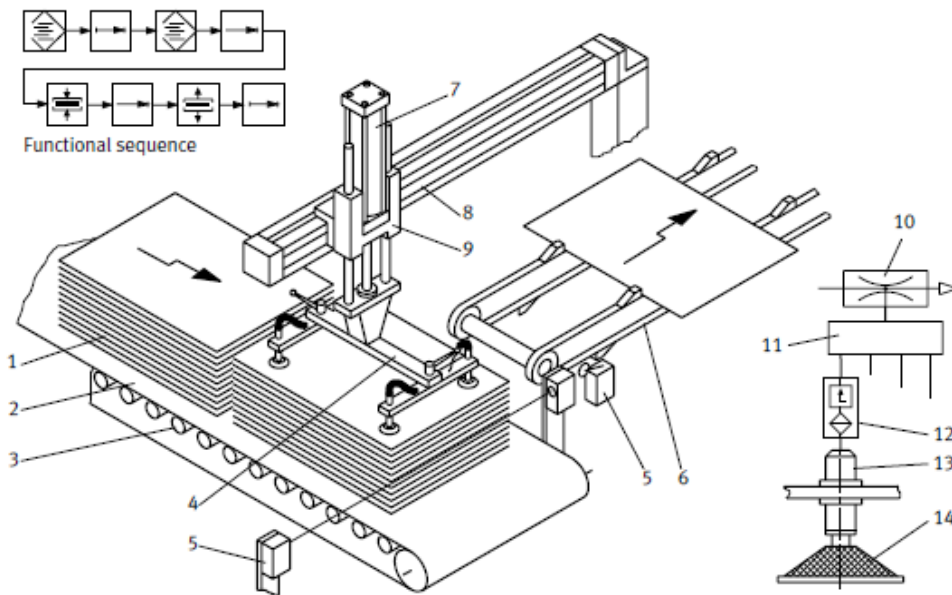
Pneumatic applications: automation and load placement.



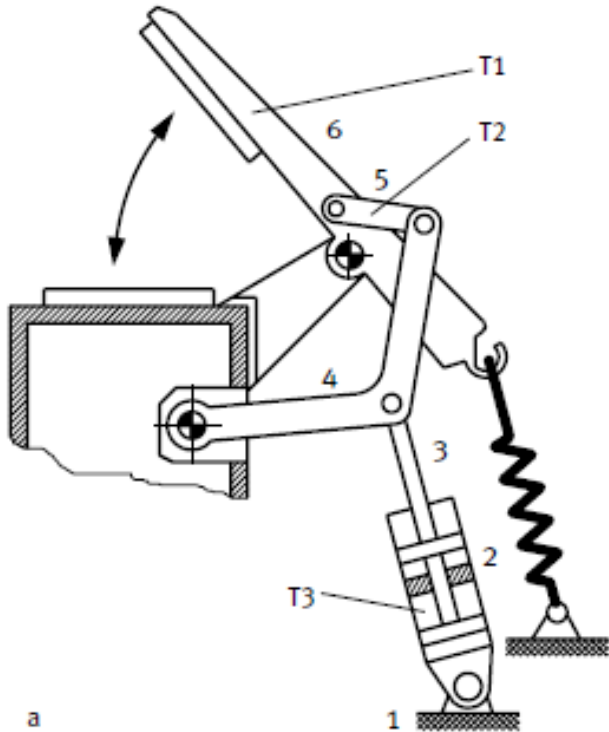
1.4. Applications.

Pneumatic applications: positioning and bottling.



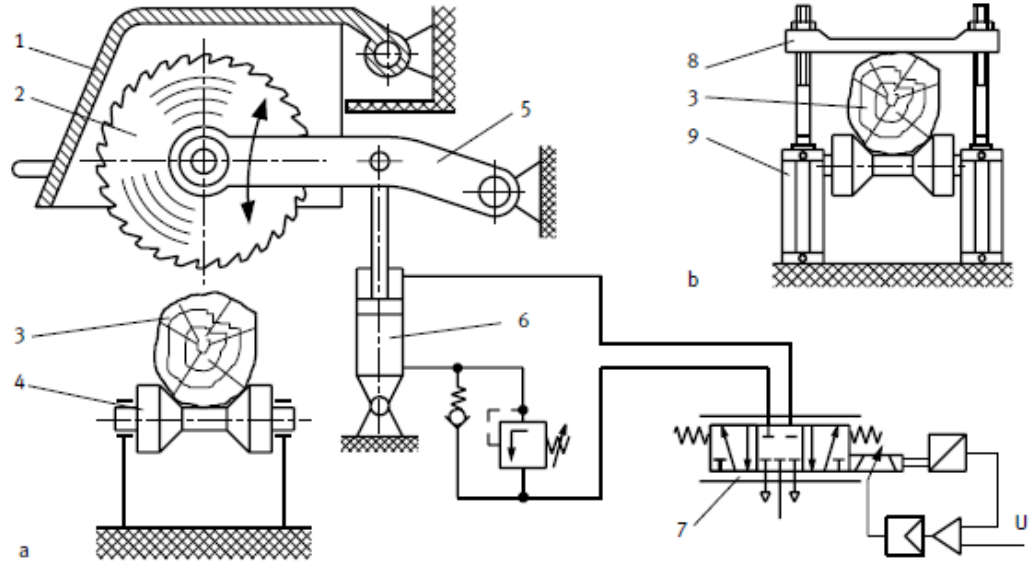


Pneumatic applications: presses.



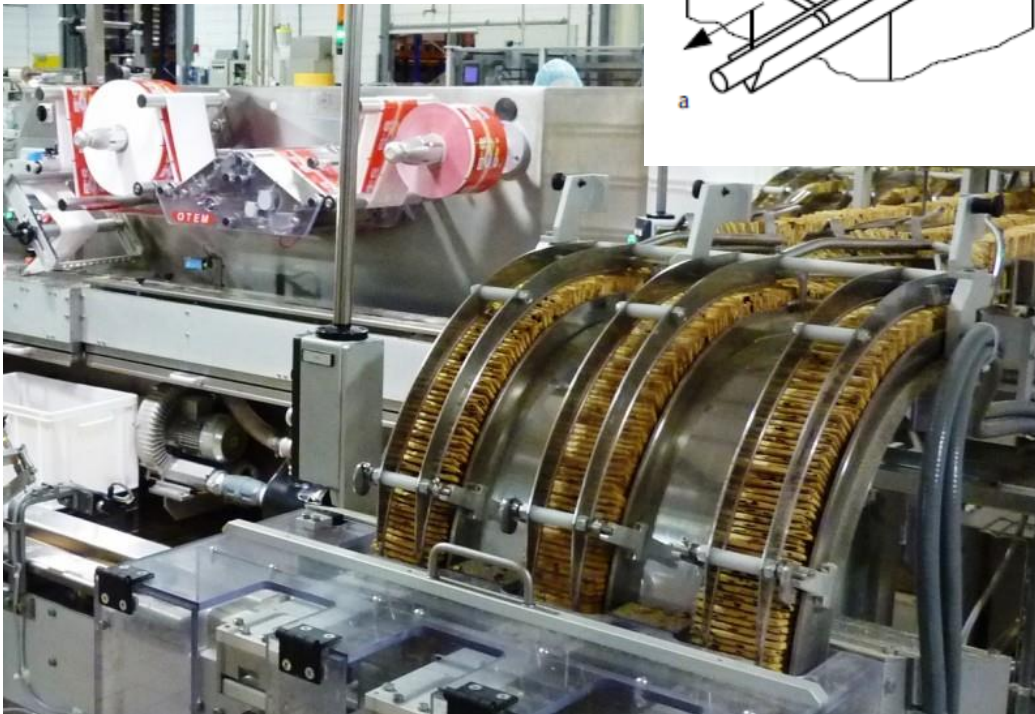
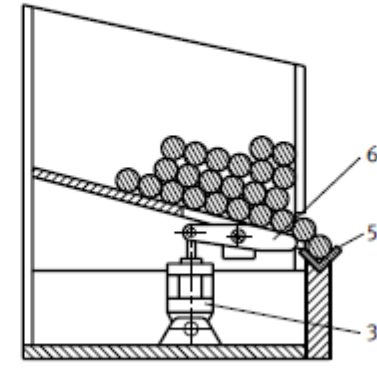
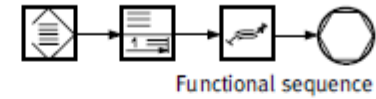
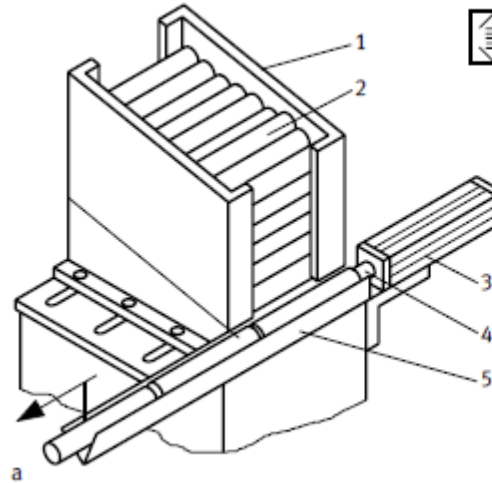
1.4. Applications.

Pneumatic applications: saw.



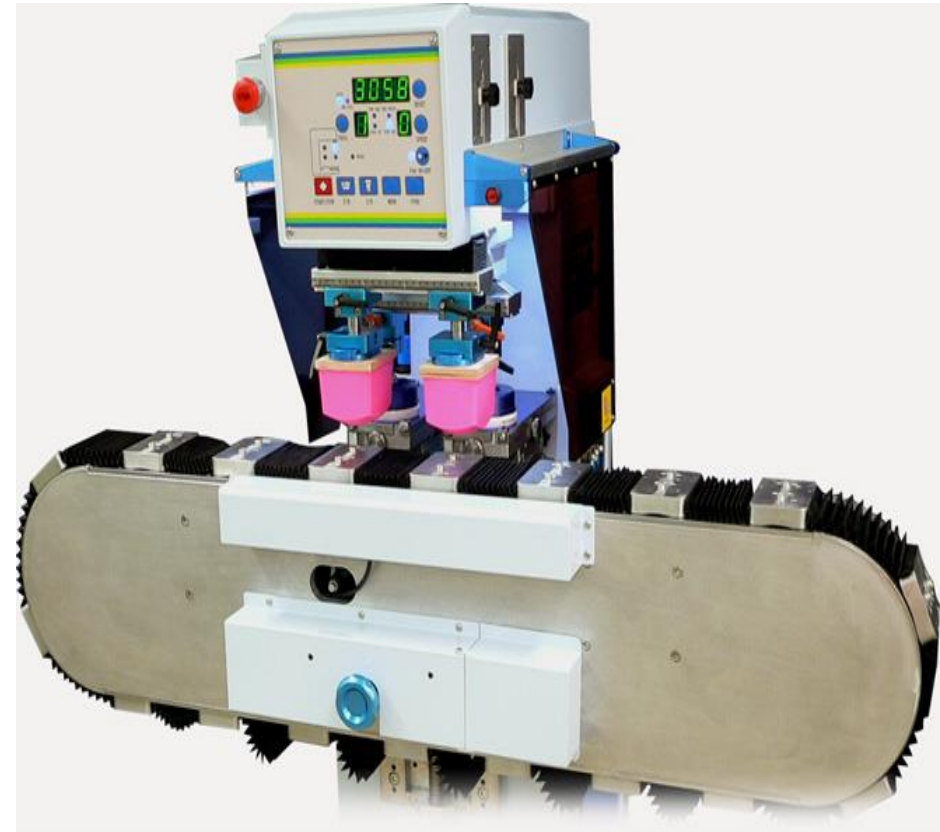
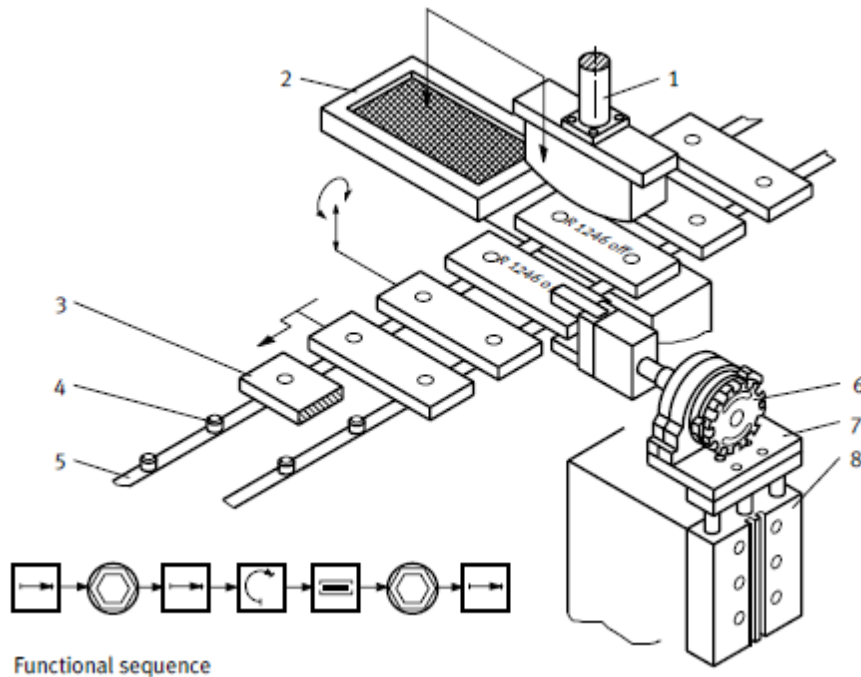
1.4. Applications.

Pneumatic applications: feeder.



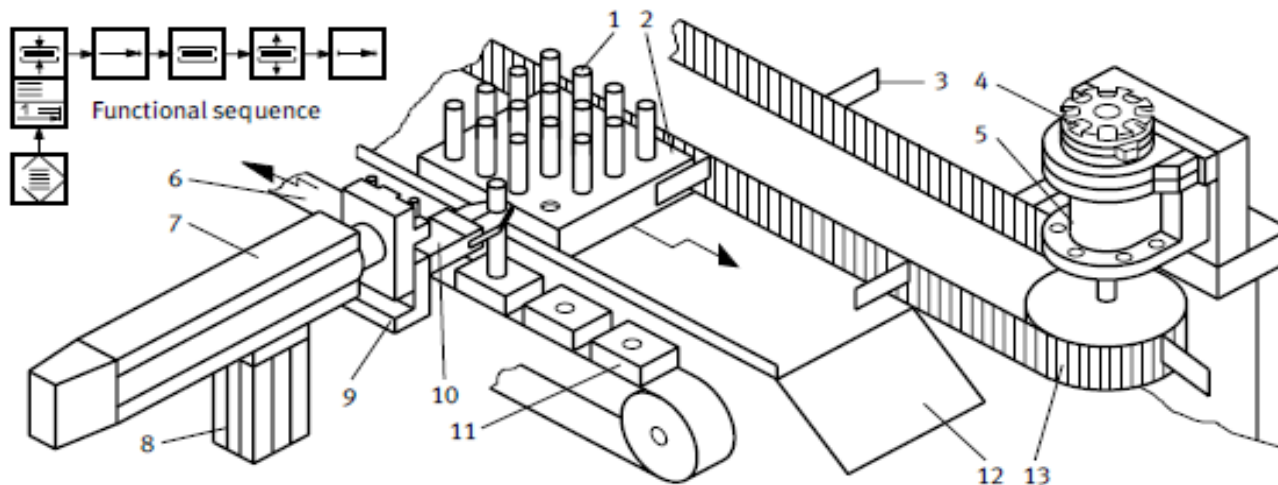
1.4. Applications.

Pneumatic applications: industrial printing.



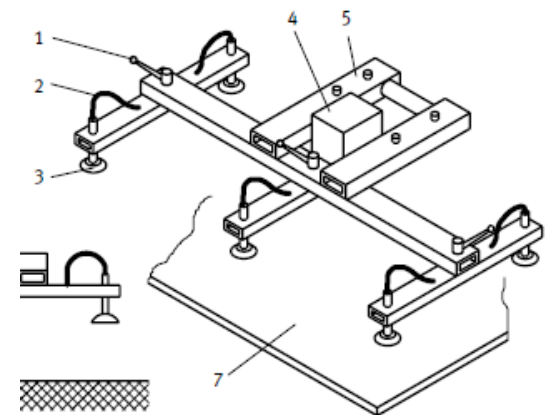
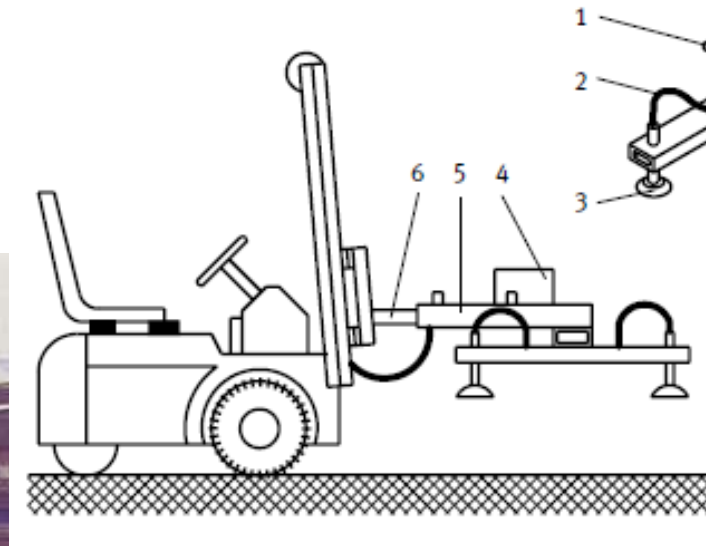
1.4. Applications.

Pneumatic applications: placing and positioning.



1.4. Applications.

Pneumatic applications: transportation.



1.5. Symbols.

CETOP standard (Commission European for Transmissions using Oleo-hydraulics and Pneumatics).

□ General facts.

□ Energy generation.

□ Pumps.

□ Motors.

□ Cylinders.

□ Converters/multipliers.

□ Distribution and control.

□ Directional valves.

□ Pressure, flow-rate valves.

□ Limiting valves/anti-return valves.

□ Energy transmission.

□ Tanks, accumulators.

□ Filters, lubricators, refrigerators.

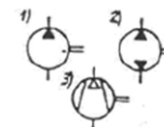
□ Valve control systems.

□ Other devices.

Constant flow rate pumps and compressor:

1) and 2) pumps, 3) compressor. 2) reversible. 1) and 3) one way.

Symbol

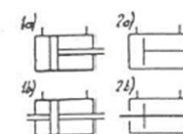


Double stroke cylinder:

The fluid pressure promotes the cylinder movement in both directions (advance and return). 1a) and 2a) single rod. 1b) and 2b) passing rod.

Symbol

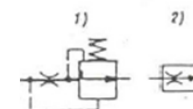
Detailed Simplified



Flow rate regulation:

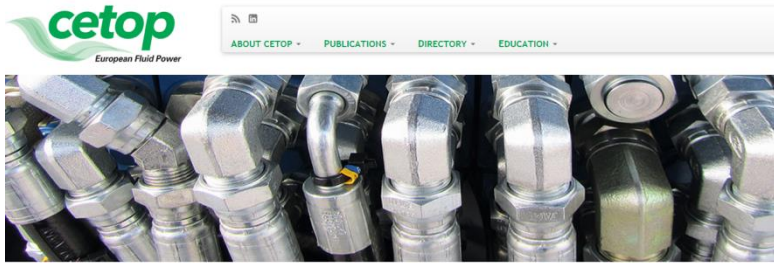
The flow rate is almost constant and independent of the pressure variations. 1) detailed, 2) simplified. When there is variable flow rate regulation, an arrow is added.

Symbol



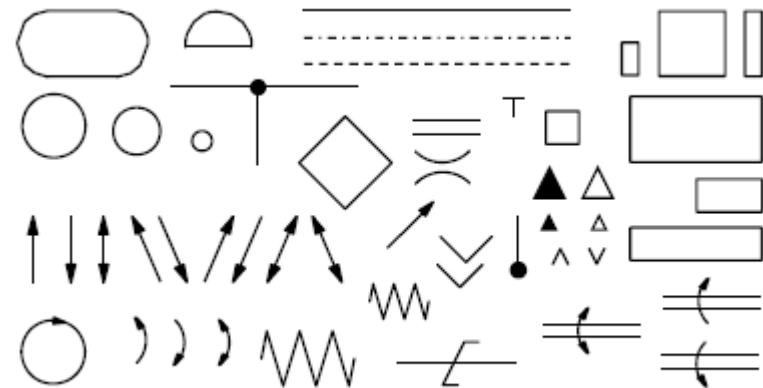
1.5. Symbols.

General symbols: pipes and connections.



- Pneumatic symbols conform to and are devised from the International Standard ISO 1219-1 1991.
- This covers graphical symbols for Fluid Power Systems and Components.
- Port markings for fluid power valves are not covered by the ISO standard. These are taken from the recommendations of CETOP RP 68 P.

(almost) all possible elementary symbols...



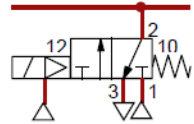
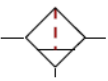
1.5. Symbols.

General symbols: pipes and connections.

❑ **Main lines:** These are the lines that are used to define the main flow rate paths on the circuit. They are drawn with continuous lines and, sometimes, can have different colours depending on the working pressure.

❑ **Pilot lines:** Indicate the auxiliary flow pass for the valve's movement or elements moved with a pressure signal. They are drawn with long dashed lines.

❑ **Bleed lines:** They are in the circuit to indicate that the flow returns to the tank or to the atmosphere. They are drawn with short dashed lines.

Line		Working line, pilot supply, return, electrical	
Dashed		Pilot control, bleed, filter	
Chain		Enclosure of two or more functions in one unit	
Line		Electrical line	

Lines and functions	
Working line	
Pilot line	
Bleed line	
Connection	
Flexible connection	
(Tee) Line connection	
Crossing lines	
Flow direction	
Tank lines	
Filling lid and air filter	
Closed path	

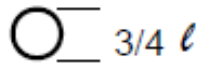
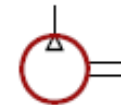
Basic Symbols (shapes)

- Symbol sets can be drawn to any size but their scale and relative proportions are determined by a basic dimension of your choice " ℓ "

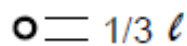
Circles



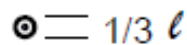
energy conversion units



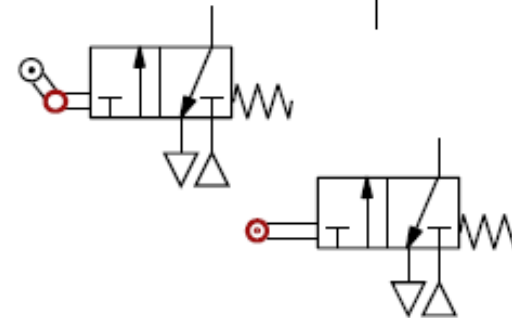
measuring instrument



mechanical link

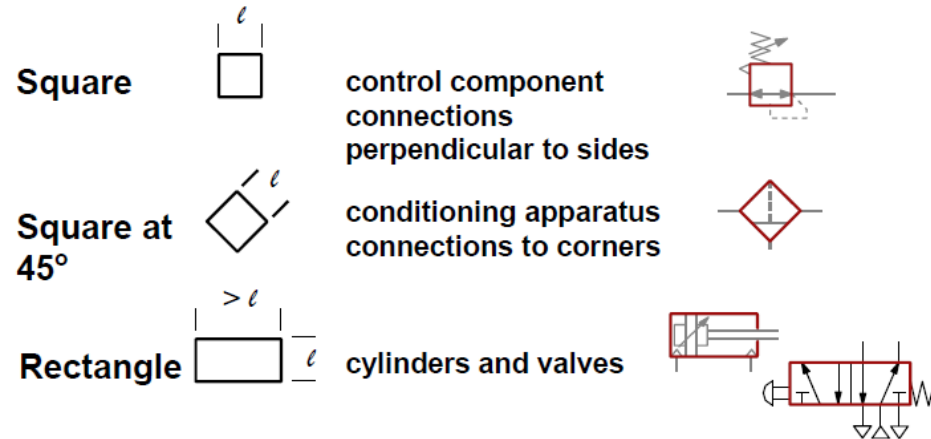


roller



1.5. Symbols.

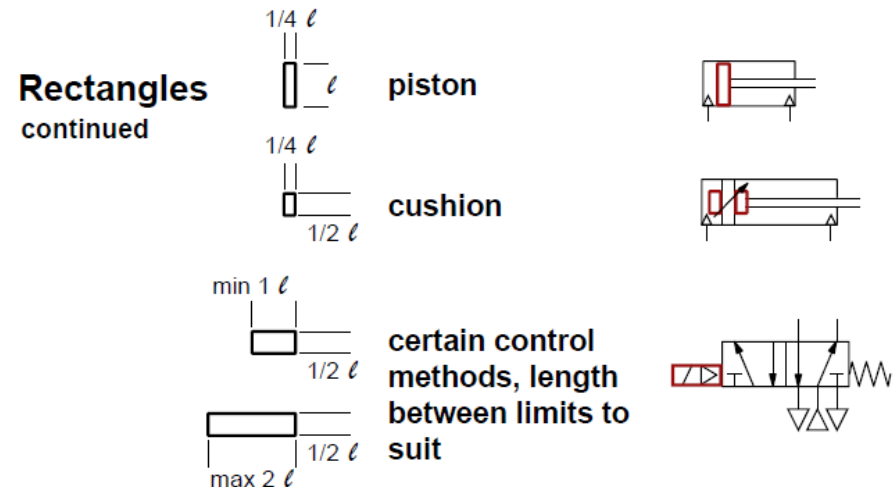
Basic Symbols (shapes)



Shapes are simple, but their combination....

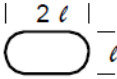
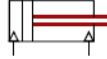
When made at hand so much care is difficult

Basic Symbols (shapes)



1.5. Symbols.

Basic Symbols

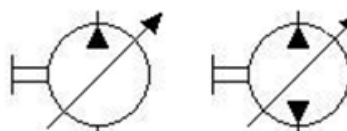
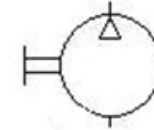
Semi-circle		rotary actuator, motor or pump with limited angle of rotation	
Capsule		pressurised reservoir air receiver, auxiliary gas bottle	
Double line		mechanical connection piston rod, lever, shaft	

1.5 Additional		
		Connection.
		Spring.
		Flow regulation.

2. Functional symbols		
2.1 Triangles		
2.1.1 Solid	 	Hydraulic fluid direction.
2.1.2 Hollow	 	Pneumatic fluid direction.
2.2 Arrows		
2.2.1 Vertical		Rotating direction, internal circulation in a valve.
2.2.2 Inclined		Possibility to regulate or change.

1.5. Symbols.

Energy generation: pumps, compressors...

3. Pumps and compressors		
3.1 Pump with fixed cubic capacity		One or two fluid directions
3.2 Variable cubic capacity pump		One or two fluid directions
3.3 Compressor		Only one flow direction



1.5. Symbols.

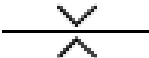
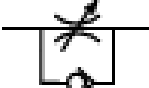
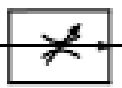
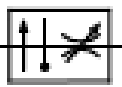
Hydraulic:

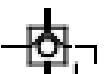
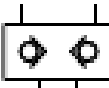
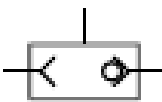
Flow directional valves

Possible valves		Possible 4/3 valves	
2/2 Valve		4/3 Electro-valve	
3/2 Valve		4/3 Electro-valve (piloted)	
4/2 Valve		Proportional valve	
4/3 Valve (i)		Servo-valve	
4/3 Valve (ii)		Proportional valve. Position detection.	
4/3 Valve (iii)		Proportional valve. Pressure transducer.	
4/3 Valve (iv)		Cartridge valves	
3/2 Valve (i)		Cartridge valve (2:1)	
3/2 Valve (ii)		Cartridge valve (14.3:1)	

1.5. Symbols.

Flow control valves.

Flow rate control	
Fixed flow control	
Calibrated hole	
Variable flow regulator, bi-directional	
Variable flow regulator, uni-directional	
Pressure regulator (Pressure compensated)	
Pressure regulator (Pressure and temperature compensated)	

Flow rate control	
Non-return valve	
Non-return valve with spring (opening)	
Non-return valve piloted	
Non-return valve piloted and with bleed line	
Double non-return valve piloted	
Line selection valve	
Non-return valve with spring (closure).	

1.5. Symbols.

Pressure valves and accessories.

Pressure valves

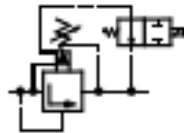
Relief valve



Piloted relief valve



Venting



Pressure regulator.
Pre-set relieving.



Pressure regulator
(3 paths)



Pressure regulator
(2 paths and drain)

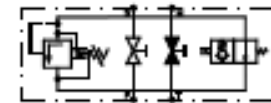


Accessories

Accumulator



Accumulator group



Thermometer



Liquid level



Quick switch



Cooling



Heater



Filter

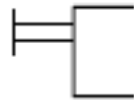




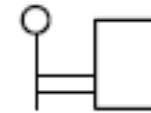
1.5. Symbols.

Hydraulic and pneumatic. Valve operation. Manual operators:

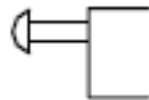
General manual



Lever



Push button



Pedal



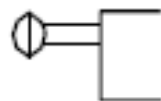
Pull button



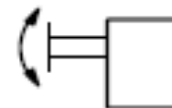
Treadle



Push/pull button



Rotary knob

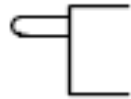




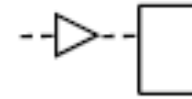
1.5. Symbols.

Hydraulic and pneumatic. Valve operation. Mechanical operators:

Plunger



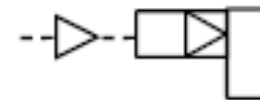
Pressure



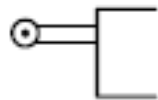
**Spring normally
as a return**



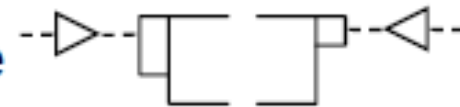
Pilot pressure



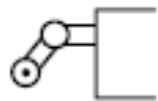
Roller



Differential pressure



**Uni-direction
or one way trip**



Detent in 3 positions





1.5. Symbols.

Hydraulic and pneumatic. Valve operation. Electrical operators:

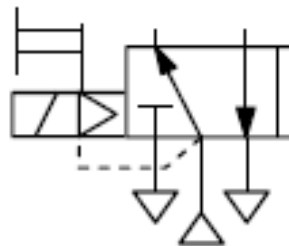
Solenoid direct



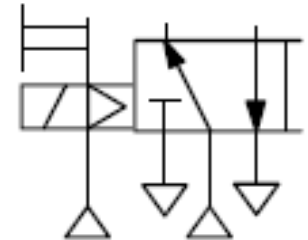
Solenoid pilot



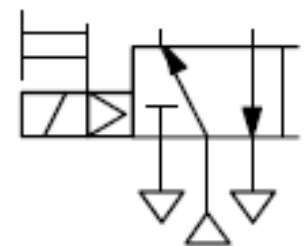
Solenoid pilot with manual override and integral pilot supply



Solenoid pilot with manual override and external pilot supply



When no integral or external pilot supply is shown it is assumed to be integral



Some are combined operators. For instance: pressure and manual.



1.5. Symbols.

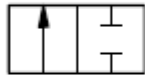
Pneumatic:

Basic valves before
operators are added

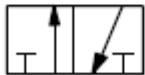
Examples, push button operated
with spring return

Number of ports Number of positions

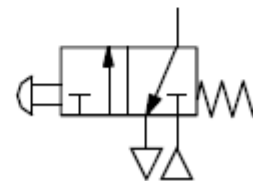
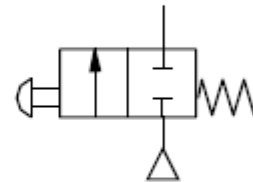
Function 2/2



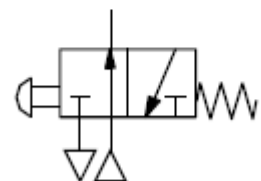
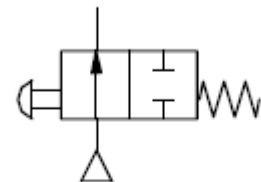
Function 3/2



Normal position



Operated position





1.5. Symbols.

Pneumatic:

Valve functions 5/3

- **Three position valves have a normal central position that is set by springs or with a manual control such as a lever**
- **The flow pattern in the centre position varies with the type. Three types will be considered**
- **1, All ports sealed**
- **2, Outlets to exhaust, supply sealed**
- **3, Supply to both outlets, exhausts sealed**

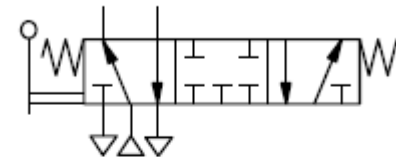
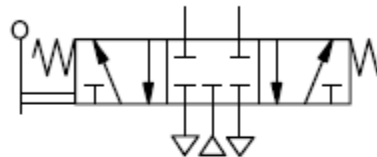


1.5. Symbols.

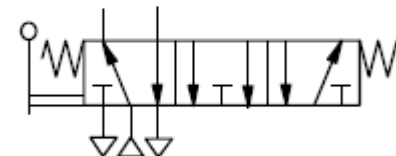
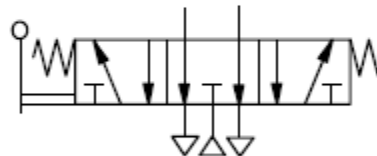
Pneumatic:

- All valves types shown in the **normal position** **first operated position**

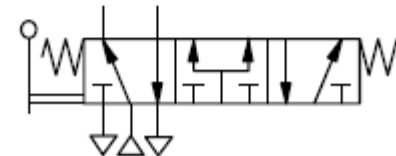
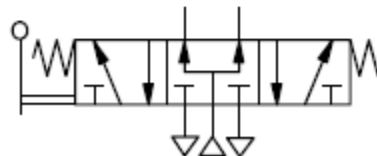
- Type 1. All ports sealed



- Type 2. Outlets to exhaust



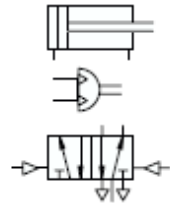
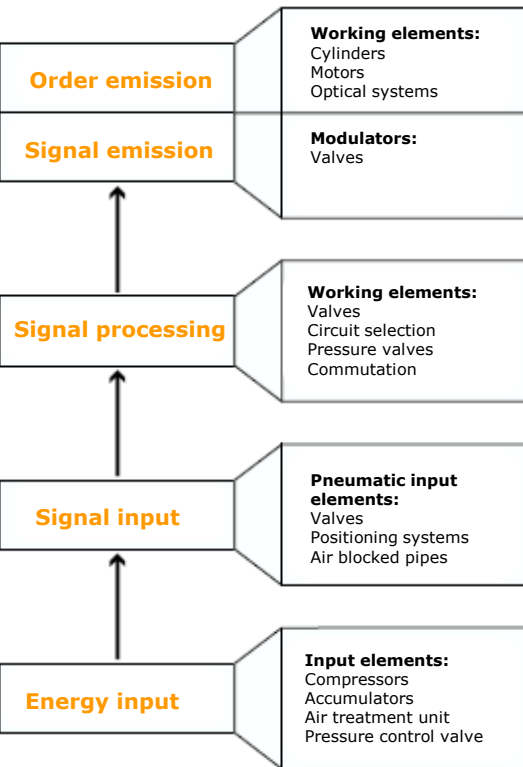
- Type 3. Supply to outlets



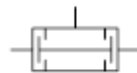
Exercise: Draw the valves in the second operated position.

1.5. Symbols.

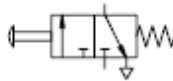
Pneumatic systems hierarchy:



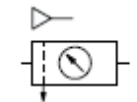
- Water separator with manual drain



- Water separator with automatic drain



- Filter with manual drain



- Filter with automatic drain

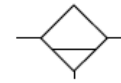
- Lubricator

- Dryer

- Cooler with and without coolant flow lines

- Heater

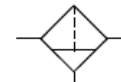
- Combined heater / cooler



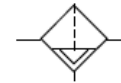
- Compressor and electric motor



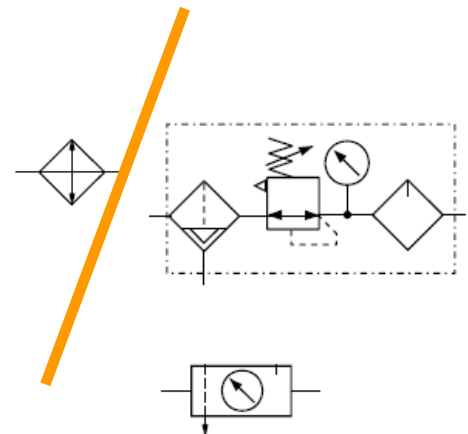
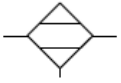
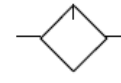
- Air receiver



- Isolating valve



- Air inlet filter

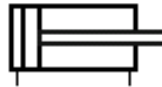


Filter Regulator Lubricator (FRL)

1.5. Symbols.

Pneumatic actuators:

double-acting cylinder
with single piston rod



double-acting cylinder
with through piston rod



differential cylinder



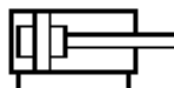
double-acting telescopic cylinder



double-acting cylinder
with single end position cushioning



double-acting cylinder
with end position cushioning at both ends



double acting cylinder
with adjustable end position cushioning
at both ends



- Single acting sprung instroked



- Single acting sprung outstroked



- Single acting sprung instroked magnetic *



- Single acting sprung outstroked magnetic *

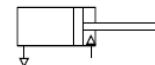


* ISO 1219-1 provides no example for magnetic cylinders

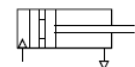
- Single acting normally instroked external force returns



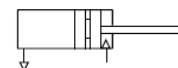
- Single acting normally outstroked external force returns



- Single acting normally instroked magnetic external force returns



- Single acting normally outstroked magnetic external force returns



Rotary actuators

- Semi rotary double acting



- Rotary motor single direction of rotation

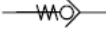


- Rotary motor bi-directional



1.5. Symbols.

Pneumatic valves:

- Non-return valve simplified 

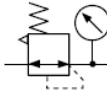
- Flow regulator uni-directional 

- Flow regulator bi-directional simplified 

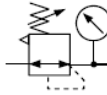
- Two pressure 'AND' simplified 

- Shuttle valve 'OR' simplified 

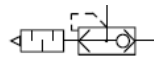
- Pre-set relieving 

- Pre-set relieving with pressure gauge 

- Adjustable relieving 

- Adjustable relieving with pressure gauge 

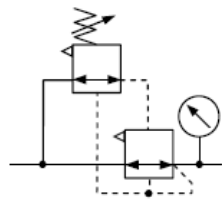
Note: The output can exhaust back and is not trapped

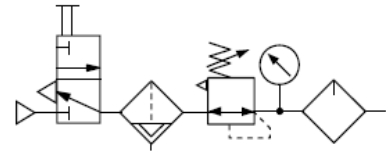
- Quick exhaust valve with silencer simplified 

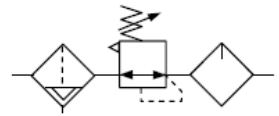
- Silencer 

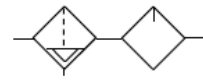
- Pilot operated 

- Pilot regulator with independent feedback 

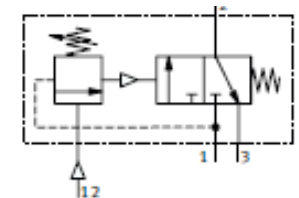
- Pilot operated regulator controlled from pilot regulator 

- FRL with shut off valve and pressure gauge 

- Lubro-control unit 

- Filter and lubricator 

Valve with pressure control





1.6. Hydraulic fluids.

Functions:

- ❑ Pressure transmission.
- ❑ Lubrication of the moving parts.
- ❑ Cooling of friction losses.
- ❑ Oscillation damping (expansions).
- ❑ Protection against corrosion.
- ❑ High sensitivity for signal transmission.

Properties:

- ❑ Compressibility (incompressible fluid).
- ❑ Thermal expansion.
- ❑ Density (preferable low).
- ❑ Viscosity (capacity to afford tangential forces).
- ❑ Unctuousity (capacity to cover surfaces).
- ❑ Demulsibility (avoid bubbling effects).
- ❑ Anticorrosive, antioxidant.
- ❑ Non toxic, resistance against flames, environmentally friendly.

Compressibility

- ❑ As any liquid, hydraulic fluids are considered **incompressible**.

Example: density variation in an oil cube of 10 cm of side and 850 kg/m³ of density, when the pressure is increased 500 bar ($K=1.4 \text{ GN/m}^2$).

$$K = \frac{dP}{d\rho/\rho} \rightarrow \rho_1 = \rho_0 \cdot \exp\left(\frac{\Delta P}{K}\right)$$

Compressibility Module

$$\rho_1 = 850 \cdot \exp\left(\frac{500 \cdot 10^5}{1.4 \cdot 10^9}\right) = 881 \frac{\text{kg}}{\text{m}^3}$$

$$\frac{\Delta\rho}{\rho} = \frac{31}{850} \cdot 100 = \boxed{3.6\%}$$



1.6. Hydraulic fluids.

Thermal Expansion

- Density variation in hydraulic fluids because of temperature changes is also very small.

Example: density variation in an oil cube of 10 cm of side and 850 kg/m³ of density, when the temperature is increased 100 K ($\alpha=0.0007 \text{ K}^{-1}$).

$$\Delta\vartheta = \alpha\vartheta_0\Delta T = 0.0007 \cdot 10^3 \cdot 100 = 70 \text{ cm}^3$$

$$m = \text{const.} \rightarrow \rho_1 = \rho_0 \frac{\vartheta_0}{\vartheta_1} = 850 \frac{1000}{1070} = 794 \frac{\text{kg}}{\text{m}^3}$$

$$\frac{\Delta\rho}{\rho} = \frac{850 - 794}{850} \cdot 100 = 6.5\%$$

Volumetric Expansion

- **Conclusion:** for practical purposes, the **density** of hydraulic fluids can be considered practically **constant** (it does not change with **P** or **T**).
- Typical density values are around **850-900 kg/m³**.
- Density is important mainly in the analysis of **transitory phenomena** (inertia).



1.6. Hydraulic fluids.

Viscosity:

- It is the most important property of the hydraulic fluids.
- It is not desirable that the viscosity is neither too high nor too low; manufacturers often include recommended values in their catalogs.

High viscosity means...

- High resistance to flow
- Head losses rise
- High power consumption
- Temperature rise (friction)

Low viscosity means...

- Leaks rise between joints
- Low lubrication due to reduction in the fluid thickness

Dynamic Viscosity (μ)

$$\left[Pa \cdot s = \frac{kg}{m \cdot s} \right] \quad (S.I.)$$

$$\left[P = \frac{g}{cm \cdot s} \right] \quad (C.G.S.)$$

Poise

$$(1 P = 0.1 Pa \cdot s)$$
$$(1 cP = 0.01 P)$$

Kinematic Viscosity (ν)

$$\nu = \frac{\mu}{\rho} \quad \left[\frac{m^2}{s} \right] \quad (S.I.)$$

$$(1 St = 10^{-4} m^2/s)$$
$$(1 cSt = 0.01 St)$$
$$\left[St = \frac{cm^2}{s} \right] \quad (C.G.S.)$$

Stoke

1.6. Hydraulic fluids.

- ❑ The **ISO 3448** standard classifies hydraulic fluids based on their viscosity in **cSt** at reference temperature of **40 °C**. The most common viscosity grades are: 5, 10, 22, 32, 46, 68, 100, 150, 220 and 320
- ❑ There are also other measurements and classifications that are more obsolete or used in specific applications, such as the Redwood or Saybolt seconds (**SUS**) or the classifications of the **SAE** (Society of Automotive Engineers), **AGMA** (American Gear Manufacturers Association) or the **API** (American Petroleum Institute).

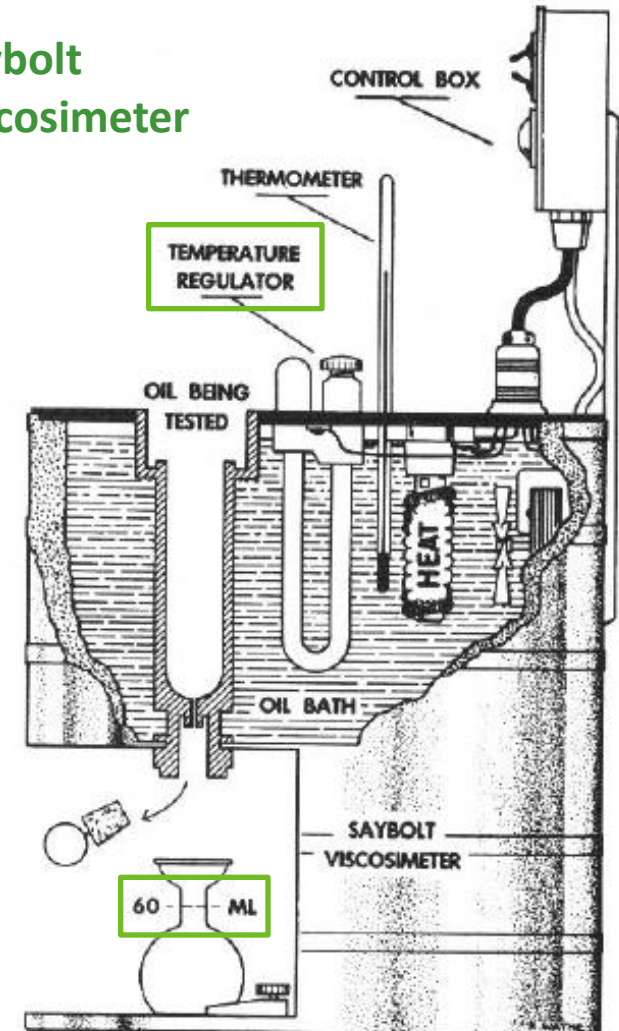
Equivalence SUS - cSt

$$v(cSt) = 0.226 \cdot t - \frac{195}{t}, t \leq 100 \text{ SUS}$$

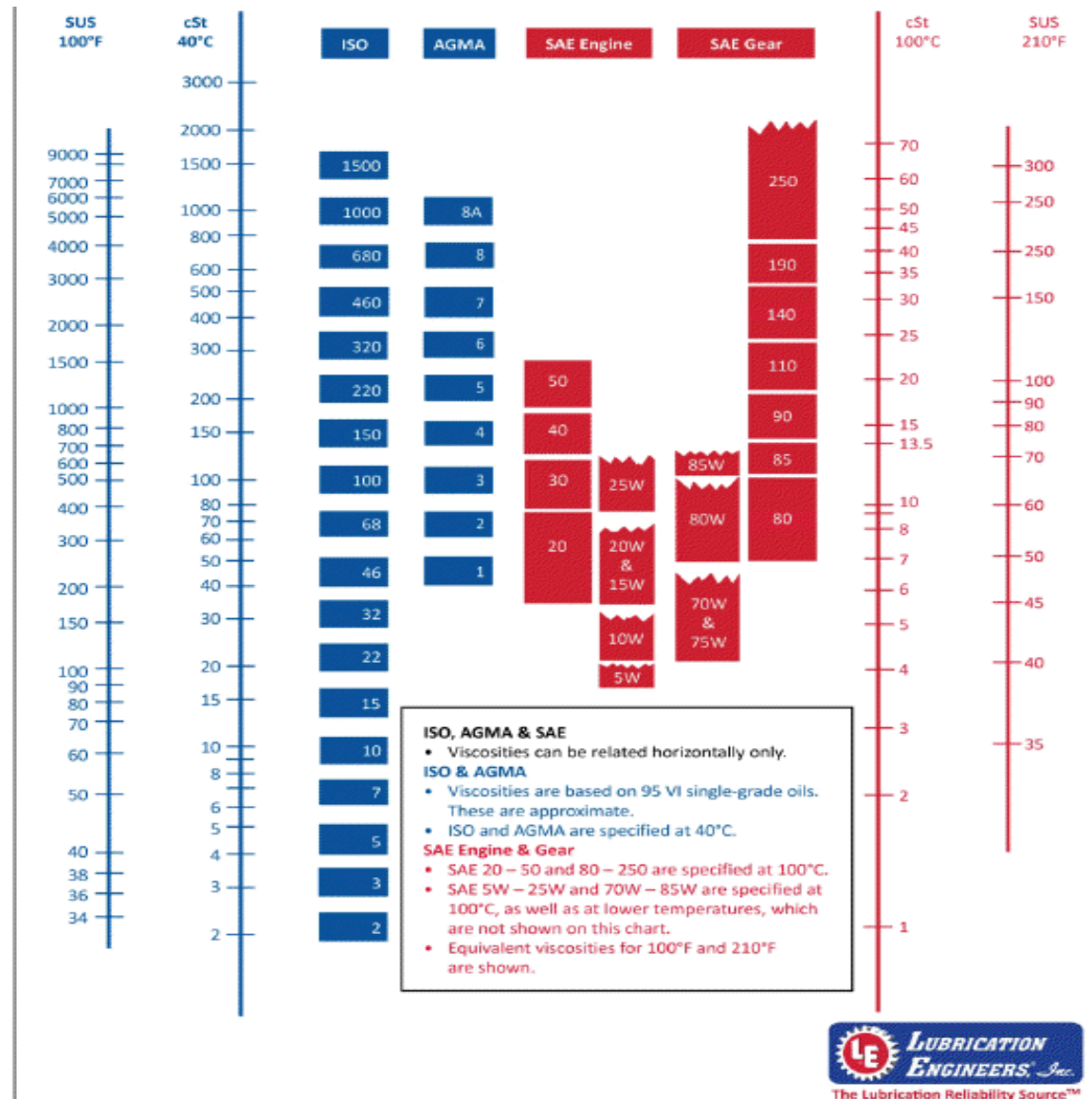
$$v(cSt) = 0.220 \cdot t - \frac{135}{t}, t > 100 \text{ SUS}$$

100 F (40 °C) or 210 F (100 °C)

Saybolt Viscosimeter



1.6. Hydraulic fluids.





1.6. Hydraulic fluids.

Viscosity Index (VI):

- ❑ Measures the change in viscosity due to a temperature variation.
- ❑ Important in external systems under high temperature changes.
- ❑ Follows the standard **ASTM-D-2270-93**:

$$VI = \frac{L - U}{L - H} \times 100$$

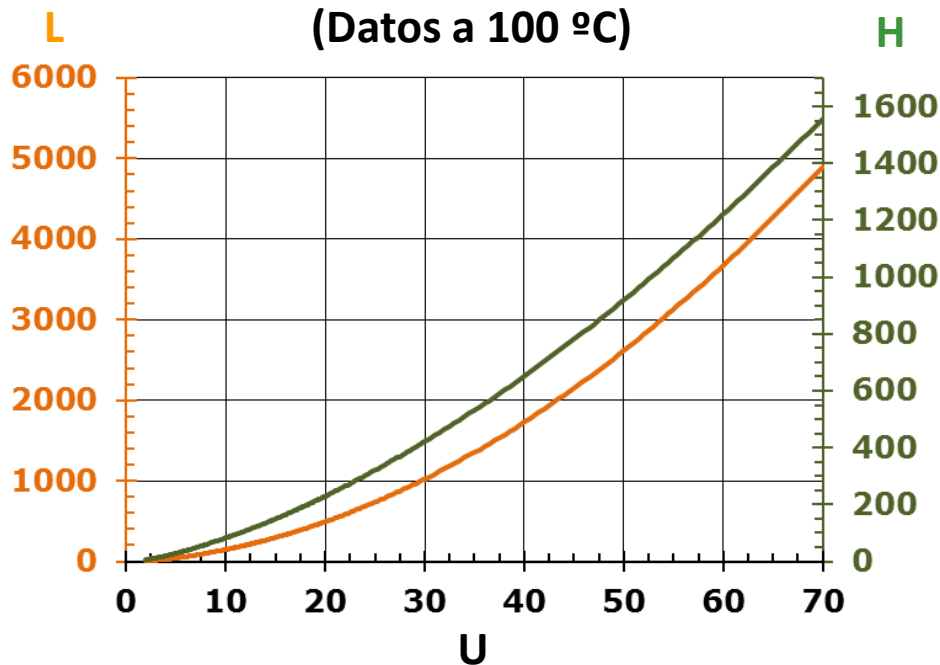
Low VI → High viscosity variation with T

High VI → Stable viscosity

- **U**: viscosity (cSt) at 40°C of desired fluid.
- **L**: viscosity (cSt) at 40°C of a fluid with VI=0 that at 100°C has the same viscosity as **U**.
- **H**: viscosity (cSt) at 40°C of a fluid with VI=100 that at 100°C has the same viscosity as **U**.

(Note: Due to the growing development of the Chemical Industry, it is currently possible to find hydraulic fluids with VI> 100)

1.6. Hydraulic fluids.



(From data of ASTM-D-2270-93)

Example: obtain the VI of a SAE 30 oil with SUS=538 at 40°C and SUS=68 at 100 °C.

Solution:

$$v(cSt) = 0.226 \cdot 68 - \frac{195}{68} = 12.5 \text{ at } 100^\circ\text{C}$$

$$v(cSt) = 0.220 \cdot 538 - \frac{135}{538} = 118.1 \text{ at } 40^\circ\text{C}$$

Figure



$$L = 216.6 \text{ cSt}$$
$$H = 114.7 \text{ cSt}$$

$$VI = \frac{216.6 - 118.1}{216.6 - 114.7} \times 100 = 96$$

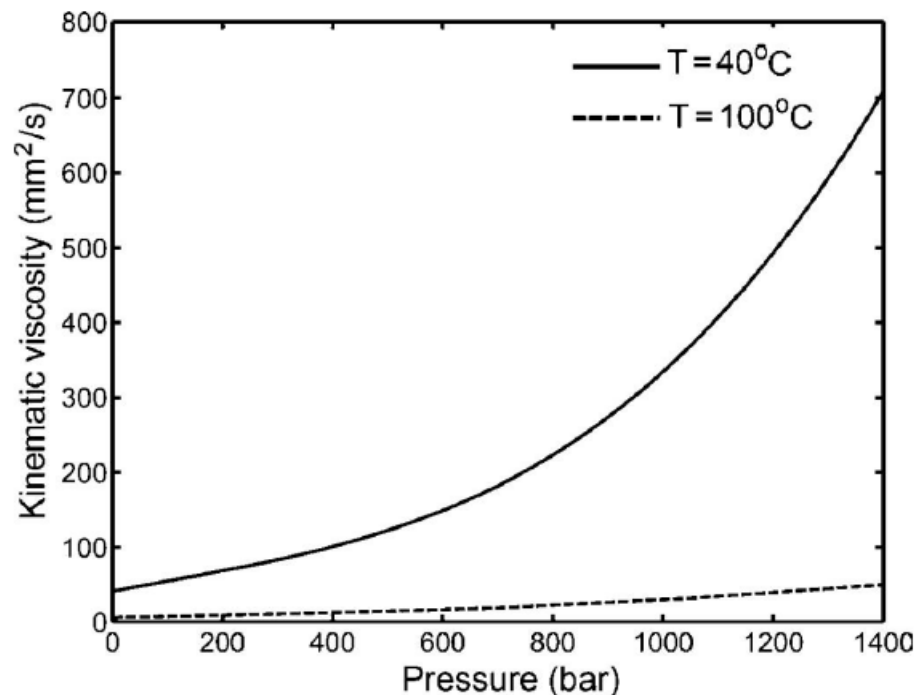


1.6. Hydraulic fluids.

- ❑ Another important property of hydraulic fluids found directly related to viscosity is the **pour point**.
- ❑ It is defined as the **lowest temperature** at which the fluid behaves as such (that is, it **is able to flow**).
- ❑ The pour point should be at least **7 °C** below the minimum expected temperature.

Pressure effect:

- Low dependence in comparison with temperature
- Viscosity rises with pressure increment
- However, **under 300 bar can be neglected**





1.6. Hydraulic fluids.

Types of Hydraulic Oils (ISO 6743/4):

HH: Mineral oil without additives	Mineral (80%)
HL: Mineral oil with anti-oxidant and anti-corrosive additives	
HM: Mineral oil of the HL type with anti-wear additives	
HV: HM type oil with additives that improve the viscosity index	

Abundant and cheap
Good lubricating properties
Compatible with most gaskets

High temperature degradation (additives)
More sensitive to viscosity variation with P
FLAMMABLE

HFAE: Oil in water emulsion (2-5% oil). Little lubrication	Water based
HFAS: Chemical Solutions	
HFB: Water in oil emulsion (40% water). More popular. Lubrication capacity is reduced. Pumps should rotate slower (and be larger)	
HFC: Polyglycol solution. Sensitive to high temperatures (especially suitable for low temperature use)	

HFDR: Based on esterphosphates	Synthetic
HFDS: Based on halogenated hydrocarbons	

Good thermal and oxidation properties
Good viscosity-temperature response
Toxicity and price

HETG: Triglycerides	Biodegradable
HEPG: Polyglycols	
HEES: Synthetic Esters	

(Note: remember that each type of oil is named followed by its viscosity value in cSt @ 40 °C. Example: HM32)

1.6. Hydraulic fluids.

Types:

	Mineral oil	Water-glycol	Emulsion Water-oil	Ester- phosphates	Ester organics	Hydro-chlorite carbides
Good joints	Viton, nitrile, polysulfide, neoprene, buna N	Buna S y N, nitrile, neoprene, butyl-viton, natural rubber	Buna S y N, nitrile, neoprene, viton, polysulphide	Viton, butyl, silicone, P.T.F.E., nylon	Neoprene, buna N, viton, silicone	Viton, silicone, Teflon
Not so good joints	Natural rubber, butyl, buna N	Polysulphide	Natural rubber, butyl	Neoprene, nitrile, buna N y S, polysulphide	Butyl	Nitrile, buna N y S, neoprene, butyl, polysulphide
Corrosive effects on the metal	None	Zinc, Cadmium, Magnesium	None	None	None	Cooper and alloys
Lubrication	Excellent	Good	Buena	Excellent	Excellent	Good
Toxicity	Non toxic	Non toxic	Non toxic	Some vapours	Non toxic	Might be toxic
Non-flammability	Poor	Good	Good	Excellent	Excellent	Excellent
Maximum operation temperature (°C)	90	50	50	90 to 140		65 to 260
Comparative cost	1	2 to 4	1.5 to 2	4 to 12		4 to 12
Relative density	1	1.25	1.2	1.3	1.06	1.65



1.6. Hydraulic fluids.

Filtering needs:

- ❑ Needed due to appearance of particles during the starting up or normal working conditions of the installation.
- ❑ The level of contamination of an oil by particles is specified according to **ISO 4406**.

<i>Filtering in μm</i>	<i>Kind of hydraulic technology used</i>
1-2	For very small particles in highly sensitive and extreme liability, similar to the ones needed in aeronautics or in laboratory tests.
2-5	For command and control systems, quite sensitive and with highly required liability with high pressure in the line, normally for planes, industrial robots and mechanical tools.
5-10	For hydraulic systems with highly required quality and liability with a foreseen long useful life of its components.
10-20	For general hydraulic systems, with mobile parts, medium pressure levels and medium sizes.
15-25	For low pressure systems in heavy load industry or for systems with a limited required useful life.
20-40	For systems with low pressure and high allowed gaps/plays.